



Review

Characterization of photoneutron fluxes emitted by electron accelerators in the 4–20 MeV range using Monte Carlo codes: A critical review

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ABSTRACT

Applications of electron accelerators range from nuclear waste package assay and security-related tasks to radiation therapy. Studies aiming at characterizing photoneutron fluxes generated by electron accelerators are usually based on Monte Carlo simulation. In this paper, we critically review the performance of Monte Carlo transport codes to simulate photoneutron fluxes emitted by electron accelerators operating between 4 and 20 MeV, typically the energy range of interest for the aforementioned applications. First, we go through the state of the art and lay the foundations of current theoretical knowledge on photoneutrons. By carrying out additional investigations, we show that contamination of photoneutron fluxes by electroneutrons is likely to lie between 0 and 2%. Second, we assess the characteristics of photoneutron fluxes emitted by tungsten or tantalum conversion targets and by heavy water or beryllium secondary targets. This characterization step is conducted with MCNP6.2, which is often considered as one of the reference Monte Carlo codes, and built around three parameters, i.e., photoneutron yield cross-sections, energy spectrum and angular distribution. In particular, we demonstrate that erroneous parameters in the nuclear data of MCNP6.2 lead to (γ, xn) cross-section threshold errors for two tungsten isotopes, i.e., ^{182}W and ^{186}W , inducing in turn a global underestimation of photoneutron production in tungsten. Furthermore, by taking an in-depth look at nuclear data libraries, we show that photoneutron yield cross-sections are sometimes poorly evaluated below 20 MeV, e.g., ^2H , ^9Be and ^{184}W . Third, thanks to vanadium and aluminium foils, we benchmark MCNP against photoneutron activation measurements conducted in the vicinity of three different electron accelerators, including a medical one. Simulation of these measurements denote a systematic underestimation trend extending from a few percent to a factor ten. Recent findings reported in the literature proved that photoneutron kinematics is implemented in MCNP with erroneous equations related to neutron inelastic scattering, causing hardening of photoneutron energy spectrum and may explain in part the discrepancies encountered in this MCNP benchmark study. Finally, in light of the three main sources of errors that potentially lead to unreliable results when simulating photoneutron fluxes with Monte Carlo codes – implementation of nuclear data, modelling of photonuclear physics and fundamental knowledge of photoneutron yield cross-sections – we issue recommendations for the code developers and users. Until further progress is made in the field of photoneutron simulation, mastering the current limitations of Monte Carlo codes could be the first milestone for their users.

1. Introduction

When electrons hit a target made of heavy atomic nuclei, high-energy photons are created by *Bremsstrahlung*. In turn, photons can be converted to photoneutrons – i.e., neutrons produced by photonuclear reactions – if the photon energy is above the threshold of the reaction of interest. A thorough knowledge of the photoneutron fluxes generated by electron accelerators is of major importance for many applications, among which: radiation therapy, border security and nuclear waste

package characterization. In the frame of radiation therapy, high-energy photon beams delivered by medical accelerators are often used to deposit gamma doses locally on tumors. However, simultaneously, photoneutrons may lead to undesired neutron doses (Ongaro et al., 2000; Huang et al., 2005; Abou-Taleb et al., 2018; Karimi et al., 2019; Rojas-Arias et al., 2020) deposited on the whole body of patients. In very different fields of applications, a linear electron accelerator (linac) enables to implement the photofission reaction – which is the basis of the active photon interrogation technique – in view of detecting Special

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Nuclear Material (SNM) such as uranium and plutonium isotopes either for cargo container security checks (Jones et al., 2005; Sari et al., 2019) or for nuclear waste package characterization studies (Lyoussi et al., 2000; Gmar et al., 2005; Carrel et al., 2010; Meleshkovskii et al., 2022). During active photon interrogation measurements, photoneutrons may represent spurious particles at the origin of unwanted neutron-induced fission reactions (Meleshkovskii et al., 2020). However, in view of enhancing characterization of nuclear waste packages, a linac can be at the center of a versatile platform enabling to perform either simultaneous active photon interrogation (photofission) and active neutron interrogation (fission) (Franks et al., 1982; Jallu et al., 2000) or non-simultaneous interrogations (Sari et al., 2013a). It should be nevertheless stressed that the second of these two methods enables to optimize both the high-energy photon and the photoneutron beams, and facilitates interpretation of results obtained with the two interrogation techniques. Moreover, a significant interest can also be reported towards linac-based neutron sources for a large variety of uses, such as: material analysis by neutron activation for different industrial applications (Sari et al., 2020a, 2020b; De Stefano et al., 2022), Boron Neutron Capture Therapy (BNCT) (Moss, 2014) and applications related to nuclear reactors (Fynan et al., 2021). Furthermore, one would notice that such linac-based platforms also enable to deploy a third measurement technique – i.e., high-energy imaging – which provides valuable information for the interpretation of measurement results, e.g., for the analysis of a suspicious cargo container by photofission (Sari et al., 2019) or the characterization of a nuclear waste package by active photon interrogation (Carrel et al., 2014).

The conversion targets of electron accelerators are usually made of tungsten or tantalum. It is worth to recall that the first characterization studies of photon beams and photoneutron fluxes emitted by such targets were conducted in the 1970's (Berger and Seltzer, 1970; Seltzer and Berger, 1973). The high-energy photon spectra produced by the *Bremsstrahlung* phenomenon due to electrons bombarding the target of an accelerator are continuous and have end-points energies corresponding to the electron energy. Considering electron accelerators operating in the 4–20 MeV range, photoneutrons may be produced by different materials. Indeed, the energy thresholds of the (γ, n) reactions for ^{181}Ta or for the five isotopes of tungsten are roughly around 7 MeV, i.e., the neutron separation energy of the emitter nuclei. Therefore, if the electron energy is above this threshold, photoneutrons will be directly produced by the target of an electron accelerator. Moreover, if the electron energy is also above the $(\gamma, 2n)$ reaction threshold, these reactions may also be brought into play and will contribute to the production of photoneutrons. Nevertheless, when the electron energies are close or below the (γ, n) reaction thresholds of the target materials, secondary targets including nuclei with lower (γ, n) energy thresholds can be used to achieve a significant photoneutron yield. The lowest energy threshold of the (γ, n) reaction is for ^9Be with a value of 1.67 MeV (IAEA, 2000). Another candidate would be deuterium (D or ^2H) with a (γ, n) reaction energy threshold at 2.22 MeV (IAEA, 2000), which is more easily available in the chemical form of heavy water (deuterium oxide, D_2O). Secondary targets made of beryllium or heavy water have been commonly used with low energy electron accelerators operated as low as 4 MeV (Lakosi et al., 2008; Lakosi and Nguyen, 2008). However, as beryllium is toxic, a secondary target made of heavy water is often preferred, although tritium (T or ^3H) is potentially produced during the irradiation process. Considering x the neutron multiplicity equal to 1 or 2, Table 1 gathers weight percentages and energy thresholds of (γ, n) and $(\gamma, 2n)$ reactions for each isotope (IAEA, 2000) of the most common photon-to-neutron conversion targets found in electron accelerators operating in the 4–20 MeV range. For a given electron energy in this range, and depending on the use of one or two targets, their compositions and dimensions, and on the settings of the accelerator, the photoneutron emission intensity begins around 10^7 and reaches 10^{11} neutrons per second in 4π sr (Jallu et al., 1999; Chakhlov et al., 1999; Eshwarappa et al., 2007; Yang et al., 2007; Sari et al., 2013b).

Table 1

The most common photon-to-neutron conversion targets of electron accelerators operating in the 4–20 MeV range: weight percentages and energy thresholds of (γ, n) and $(\gamma, 2n)$ reactions per isotope (IAEA, 2000).

Conversion target	Isotope	Weight percentage per isotope (%)	Energy threshold (MeV)	
			(γ, n) reaction	$(\gamma, 2n)$ reaction
Heavy water	^2H	20	2.22	–
Beryllium	^9Be	100	1.67	20.56
Tantalum	^{181}Ta	99.99	7.58	14.22
Tungsten	^{180}W	0.12	8.41	15.35
	^{182}W	26.30	8.07	14.75
	^{183}W	14.28	6.19	14.26
	^{184}W	30.70	7.41	13.60
	^{186}W	28.60	7.19	12.95
	^{206}Pb	24.10	8.09	14.82
Lead	^{207}Pb	22.10	6.74	14.83
	^{208}Pb	52.40	7.37	14.11

One would notice that data for ^{204}Pb , which represents 1.40% of the mass of natural lead, is not provided in Table 1 and in (IAEA, 2000). Nevertheless, it can be emphasized that lead – with a melting point at 327.5°C – may not be the most common option for use as primary target of an electron accelerator. On another note, this material is often used to collimate photon beams and notably in medical accelerators. Thus, for electron accelerators devoted to radiation therapy, tungsten and lead usually account for most of the photoneutron emission (Naseri and Mesbahi, 2010; Sari et al., 2013c). Moreover, while the contribution from the conversion target may be limited, the jaws are usually the main source of photoneutrons in the case of medical accelerators, see Table IV and Fig. 10 from (Sari et al., 2013c). More broadly, neutron contamination from medical accelerators has been studied for many years in the field of radiation therapy (NCRP, 1984; Loi et al., 2006). However, recent investigations prove that characterization of photoneutron fluxes emitted by accelerators is still a topic of major interest, and even at electron energies as low as 6 MeV (Sohrabi and Hakimi, 2017). It is also noteworthy to mention that the knowledge of the (γ, n) reaction energy thresholds may be discussed considering that low (γ, n) energy threshold-elements could be found in small quantities in the materials making up accelerators, their surroundings, or even in the human body for the case of deuterium for instance. Whatever the field of application considered, Monte Carlo codes are widely used to simulate the production of photoneutrons from electron accelerators. Nevertheless, accuracy of results obtained may be the subject of questions and discussions.

In this paper, we investigate with a new eye the reliability of Monte Carlo transport codes to characterize photoneutron fluxes emitted by electron accelerators operating in the 4–20 MeV range. First, we review the state of the art and the current theoretical knowledge regarding photoneutrons. Photonuclear reactions will be described as well as emission of photoneutrons. The proportion of electroneutrons in photoneutron fluxes will be the subject of particular attention and additional estimates. Second, we evaluate the characteristics of photoneutron fluxes emitted either by tungsten or tantalum targets and then by heavy water or beryllium secondary targets. This characterization step will be conducted using MCNP6.2 (Werner, 2017, 2018), which is often considered as one of the Monte Carlo codes of reference, and according to three main parameters, i.e., photoneutron yield cross-sections, energy spectrum and angular distribution. Third, we benchmark MCNP against neutron activation measurements conducted in the vicinity of three different electron accelerators, including a medical one. Finally, we will sum up the main sources of errors that potentially lead to unreliable simulated photoneutron fluxes, and make recommendations for Monte Carlo code developers as well as users who are working with photoneutrons on purpose or affected by their undesired effects, and most likely needing to simulate their production,

characterize their fluxes, probe their impact or support interpretation of experimental data.

2. The physics of photoneutrons

2.1. Photonuclear reactions

A pioneer experiment from [Chadwick and Goldhaber \(1934\)](#) showed that deuterium exposed to a radiothorium source emitting 2.6 MeV photons produces neutrons through the following reaction: ${}^2_1\text{D} + h\nu \rightarrow {}^1_1\text{H} + {}^1_0\text{n}$. Shortly after, [Szilard and Chalmers \(1934\)](#) observed that, under the influence of radium gamma-rays, neutrons are liberated from beryllium. These findings will pave the way for a new field of research devoted to the study of interactions between photons and atomic nuclei, i.e., photonuclear physics. Such interactions can lead to the emission of different types of photoparticles, among which neutrons and protons, respectively referred to as “photoneutrons” and “photoprotons”. Nevertheless, it is important to underline that in the photon energy range considered in this study, i.e., below 20 MeV, the number of photoprotons produced depends on the atomic mass number. Indeed, the probability that a charged particle succeeds in crossing the Coulomb barrier of a heavy nucleus is low ([Courant, 1951](#)). This effect does not necessarily translate into higher energy thresholds for (γ, p) reactions than for (γ, n) reactions, as the two energy thresholds are rather close for heavy nuclei like ${}^{181}\text{Ta}$ or the five tungsten isotopes ([IAEA, 2000](#)), but into a significant difference in their cross-section values. In other words, generation of photoprotons and other charged particles by electron accelerators in the 4–20 MeV range is negligible for the case of tantalum and tungsten targets and potentially significant when using light nuclei converters composed of deuterium or beryllium.

The total photonuclear cross-section per nucleon as a function of the photon energy is shown in [Fig. 1 \(Findlay, 1990\)](#). For an incident photon energy of a few MeV and up to 20 MeV, the *Giant Dipole Resonance* (GDR) is the main mechanism responsible for the creation of photoneutrons. Thus, as in the framework of this paper we are interested in electron accelerators operating in the 4–20 MeV range, photoneutrons will mainly be produced by GDR. This model and a simple description of it was provided by [Goldhaber and Teller \(1948\)](#). During this resonance, the incident photon interacts with the nucleus as a whole, with all

neutrons and all protons moving in opposite directions within the nucleus (this motion is referred to as the “dipole vibration”). It is worth to highlight that for the case of actinides, e.g., isotopes of uranium and plutonium, GDR may lead to photon-induced fission of the nuclei (photofission), a reaction predicted by [Bohr and Wheeler \(1939\)](#) and observed experimentally by [Haxby et al. \(1940\)](#) in the *Westinghouse Research Laboratories*. Years later ([Swanson, 1979](#)), it was known that the GDR cross-section is characterized by a large peak at photon energies of approximately 20–23 MeV for light nuclei ($A \lesssim 40$) and 13–18 MeV for medium and heavy nuclei ($A \gtrsim 40$). For $A \gtrsim 40$, the energy of the peak can be approximated by $k_0 = 80 A^{-1/3}$ MeV. The width of the GDR peak varies between approximately 3 MeV for heavy nuclei and 10 MeV for light nuclei.

2.2. Emission of photoneutrons

Photoneutrons created through GDR may be emitted by two different processes. [Weisskopf \(1937\)](#) introduced the *evaporation process* to describe the expulsion of particles from highly excited heavy nuclei. [Jensen \(1948\)](#) and [Courant \(1951\)](#) postulated the existence of a *direct emission process* in which the photon interacts directly with one of the particles in the nucleus without the formation of an intermediate “compound nucleus” state. [Wilkinson \(1956\)](#) described the “resonance direct” process by which a nucleon may be emitted from the nucleus in a single-particle state.

Photoneutrons emitted by evaporation process are foreseen to be characterized by an isotropic distribution ([Courant, 1951](#)). Indeed, first, a collision between a photon of sufficient energy and a nucleus may lead to the formation of a compound nucleus in a well-defined state in which the incident energy is shared among all the constituents ([Weisskopf, 1937](#)). Second, the compound system disintegrates, which can be interpreted as if the memory of the angle of the incident photon has been lost. However, the angular distribution of photoneutrons emitted by direct emission process should be in the form of $A + B \sin^2(\theta)$ ([Courant, 1951](#)), where θ is the angle between the incoming photon and the photoneutron produced. In other words, the direct emission of a photoneutron is expected to occur at 90° from the angle of the incident photon. In summary, [Courant \(1951\)](#) predicted that the photoneutron spectrum will contain:

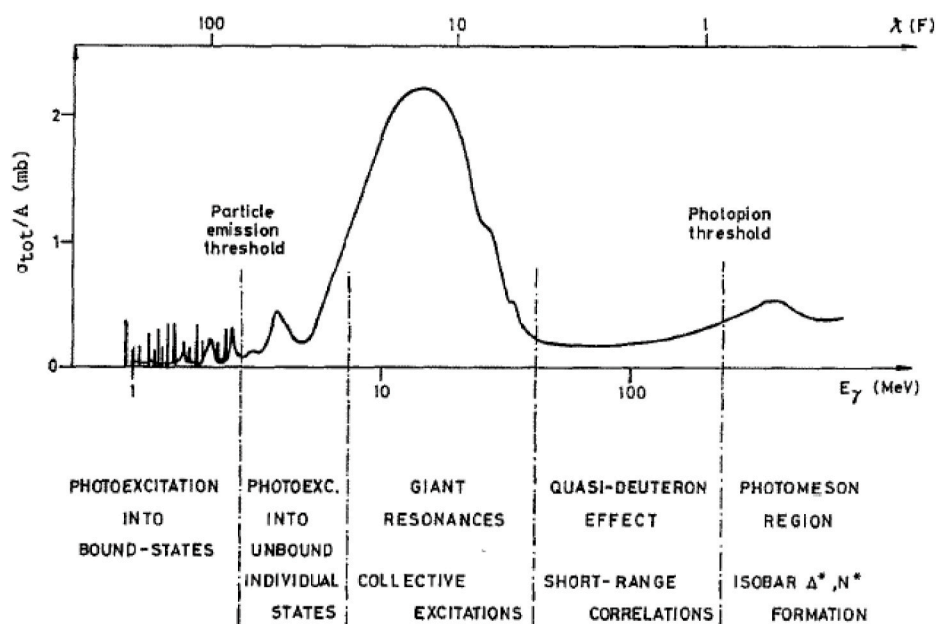


Fig. 1. Diagram of the total photonuclear cross-section per nucleon in mb as a function of the photon energy in MeV (primary x-axis) or its reduced wavelength in fm (secondary x-axis) ([Findlay, 1990](#)).

- a predominant component composed of low energy photoneutrons produced by evaporation process, following a Maxwell distribution with a most probable energy of a few MeV and characterized by an isotropic angular distribution;
- a high-energy “tail” corresponding to photoneutrons produced by direct emission process, representing a few percents of all the photoneutrons emitted and characterized by an anisotropic angular distribution.

Tagliabue and Goldemberg (1961) studied and compared the results from previous investigations conducted by numerous authors on the angular distribution of fast photoneutrons. They reported that the experiments are difficult to compare due to the diversity of experimental conditions, and that one finds some discrepancies among the authors. They also performed measurements of angular distributions of photoneutrons produced by thirteen elements. They observed a maximum at 90° in the angular distribution measurements of fast photoneutrons produced by bismuth, lead and gold. Mutchler (1966) reported in his doctoral thesis extensive experimental investigations on the angular distributions and energy spectra of photoneutrons from heavy elements. He confirmed that the photoneutron spectra are characterized by an isotropic evaporation component, which dominates at low neutron energies, and an anisotropic resonance direct component, which dominates at high neutron energies. He also claimed that the transition from one component to the other is quite abrupt, typically taking place in a photoneutron energy interval of about one to two MeV. Moreover, he estimated the fraction of directly emitted photoneutrons – defined as the integral of the direct spectrum divided by the integral of the total spectrum – to be about 14% and approximately constant with A, the mass number. Depending on the references and the radionuclides considered, the transition between the two photoneutron families, i.e., evaporation and direct emission, is essentially within the 3–5 MeV range of the total photoneutron energy spectrum.

The energy distribution of photoneutrons emitted by the evaporation process can be described by a Maxwell distribution (Blatt and Weisskopf, 1952) (page 368) (Swanson, 1979) (page 71), in which T represents the “nuclear temperature”, i.e., the temperature of the residual nucleus after emission

$$\left(\frac{dN_n}{dE}\right)_{\text{evap}} = \frac{E}{T^2} \exp\left[-\frac{E}{T}\right]. \quad (1)$$

Tosi et al. (1991) proposed an approximation for the energy distribution of photoneutrons emitted by the direct emission process, in which E_{max} is the maximum energy of the *Bremsstrahlung* photon spectrum and S_n the neutron separation energy

$$\left(\frac{dN_n}{dE}\right)_{\text{direct}} = K \ln\left[\frac{E_{\text{max}}}{E + S_n}\right], \quad (2)$$

with K a normalization factor

$$K \int_0^{E_{\text{max}} - S_n} \ln\left[\frac{E_{\text{max}}}{E + S_n}\right] dE = 1. \quad (3)$$

Noting respectively α and β the proportions of evaporation and direct photoneutrons, the total energy distribution can be calculated

$$\left(\frac{dN_n}{dE}\right)_{\text{total}} = \alpha \left(\frac{dN_n}{dE}\right)_{\text{evap}} + \beta \left(\frac{dN_n}{dE}\right)_{\text{direct}}. \quad (4)$$

Replacing the energy distributions of evaporation and direct photoneutrons by their expressions, the total energy distribution becomes

$$\left(\frac{dN_n}{dE}\right)_{\text{total}} = \alpha \left\{ \frac{E}{T^2} \exp\left[-\frac{E}{T}\right] \right\} + \beta \left\{ K \ln\left[\frac{E_{\text{max}}}{E + S_n}\right] \right\}. \quad (5)$$

Finally, replacing K by its expression, we find the total energy spectrum of photoneutrons derived by Tosi et al. (1991)

$$\left(\frac{dN_n}{dE}\right)_{\text{total}} = \alpha \frac{E}{T^2} \exp\left[-\frac{E}{T}\right] + \beta \frac{\ln\left[\frac{E_{\text{max}}}{E + S_n}\right]}{\int_0^{E_{\text{max}} - S_n} \ln\left[\frac{E_{\text{max}}}{E + S_n}\right] dE}. \quad (6)$$

Eq. (6) can be computed and is plotted in Fig. 2 for the case of a tungsten target irradiated by 15 MeV electrons (contributions from evaporation and direct photoneutrons are also shown individually) taking into account:

- a nuclear temperature T of 0.5 MeV for tungsten nuclei, as originally deduced by Barrett et al., (1973) and which is in line with more recent investigations from von Egidy and Bucurescu (2005);
- a neutron separation energy S_n of 7.35 MeV (average value calculated using the (γ, n) reaction energy thresholds provided in Table 1 for the five isotopes of tungsten and weighted by their abundancies);
- a maximum energy of the *Bremsstrahlung* photon spectrum E_{max} of 15 MeV (given the 15 MeV monoenergetic electron source);
- a fraction of direct photoneutrons β of 14%, as estimated by Mutchler (1966), and thus a fraction of evaporation photoneutrons α of 86%;
- a derivation step of 1 keV.

In this example, the normalization factor K has a value of 0.416 MeV^{-1} . We can also highlight that the total photoneutron energy spectrum is normalized to unit, the most probable energy is 0.5 MeV (which is in fact given by the “nuclear temperature”), the mean energy is about 1 MeV, and the transition area from evaporation to direct photoneutrons is around 3 MeV. It is also interesting to underline that Eq. (6) predicts that the higher the electron energy, the more energetic will be the direct photoneutrons produced by the electron accelerator. On another note, one should keep in mind that Eq. (6) is sufficient to describe the global shape of the photoneutron energy spectra. However, the use of more sophisticated algorithms, such as the one proposed by Liu et al. (1995, 1997), would show a collapse of the direct photoneutron component in the range from 0 to ~2.5 MeV, which is in agreement with measurements from Mutchler (1966).

In summary, electron accelerators operating in the 4–20 MeV range generate two types of photoneutrons:

- *boiled off photoneutrons* emitted by an evaporation process, of low energies, and for which an isotropic angular distribution is expected;
- *knock-on photoneutrons* emitted by a direct emission process, of higher energies, and for which an anisotropic angular distribution is expected (with a privileged emission angle around 90° from the incident photon).

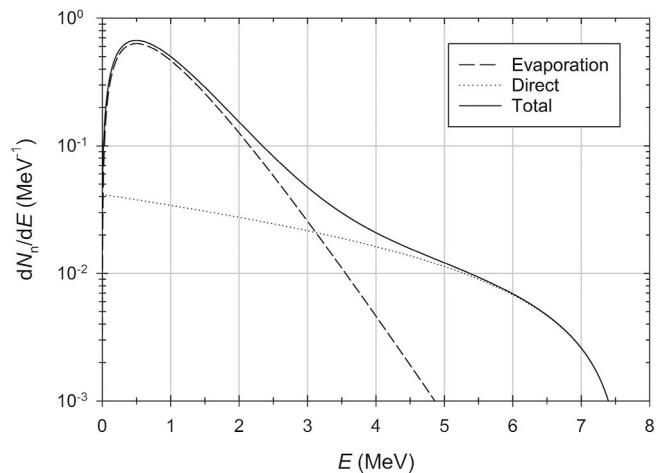


Fig. 2. Probability density as a function of the energy of photoneutrons emitted by a tungsten target irradiated by 15 MeV electrons (contributions from evaporation and direct photoneutrons are also plotted individually).

These fundamental characteristics enable to describe photoneutrons at the time of their emission at the scale of the atomic nucleus. From a macroscopic point of view, several characteristics and parameters of electron accelerators operating in the energy range studied may influence the energy spectra and angular distributions of the photoneutron fluxes produced. Indeed, it is likely that the anisotropic trend in the angular distribution of knock-on photoneutrons would be smoothed by two effects: the anisotropic angular distribution of the *Bremsstrahlung* photons; the shape, composition and density of the photoneutron-emitting material due to neutron scattering reactions. Furthermore, the isotropic trend of evaporation photoneutrons could also be affected by the geometry, composition and density of the photoneutron-emitting material. Regarding the energy of the photoneutron fluxes generated, tungsten or tantalum targets generally produce spectra characterized by a mean value of the order of 1 MeV. Nevertheless, when setups are implemented with a secondary target made of heavy water for instance (often consisting of a volume of several litres), photoneutrons emitted by deuterium nuclei could then be thermalized in the volume of the hydrogen-rich target, which results in a lower mean energy of the photoneutron flux generated. Furthermore, in the case of medical accelerators – in which photoneutrons are produced by different parts of the machine (primary target, collimators, jaws, etc.) and potentially composed of various materials – both energy spectra and angular distributions of the photoneutron fluxes are foreseen to be impacted, with noticeable differences in comparison with the characteristics of photoneutrons at the time of their emission at the scale of the atomic nucleus. Therefore, whatever the type of electron accelerators used in the 4–20 MeV range, Monte Carlo transport codes are usually necessary to characterize the photoneutron fluxes generated.

2.3. Virtual photons and electroneutrons

Fermi (1924) decomposed harmonically the electric field of a charged particle flying past an atom and compared it with the electric field of light with a proper frequency distribution. He assumed that the probability of the atom being excited or ionised by the passing particle is equal to the probability of excitation or ionisation by the equivalent radiation. Among others, he applied this assumption to the excitation due to electrons. This approximation was taken up and developed by von Weizsäcker (1934) and Williams (1935), and is commonly referred to as the *equivalent photon approximation*. Since incident electrons may produce *virtual photons* leading to electronuclear reactions, one might expect similarities between electronuclear and photonuclear reactions. Moreover, it should also be noticed that both (e, n) and (e, e' n) notations are used to refer to the electroneutron yield reactions, depending on if the outgoing electron can be detected or not during the reaction. Martins et al. (1984) measured the (e, n) cross-section for ^{63}Cu from 13.5 to 60 MeV. Wolyneć et al. (1987) measured the (e, n) cross-section for ^{181}Ta from 9 to 30 MeV. Cataldi et al. (1988) measured the electroneutron yield cross-section for ^{208}Pb , ^{209}Bi and ^{181}Ta from threshold to 22 MeV. More recently, new measurements of electroneutron yield cross-sections were reported by Yu et al. (2020) for ^{181}Ta but using a high-energy electron beam of 100 MeV, which is above the energy range studied in this paper. Nowadays, the literature is not more furnished regarding the knowledge of such cross-sections. Nevertheless, it is interesting to notice that many authors report that the electroneutron yield cross-sections are about 137 times smaller than the photoneutron yield cross-sections. This ratio, originally stated in 1984 in the NCRP report No. 79 (page 16) (NCRP, 1984), was unfortunately postulated without demonstration, and one would expect that the ratio between electronuclear and photonuclear cross-sections is not constant with energy.

In the electron energy range of interest in this paper, let us define the electronuclear yield cross-section

$$\sigma(e, xn) = \sigma(e, n) + 2 \sigma(e, 2n) \quad (7)$$

and the photoneutron yield cross-section

$$\sigma(\gamma, xn) = \sigma(\gamma, n) + 2 \sigma(\gamma, 2n), \quad (8)$$

in which the neutron multiplicities are included for both the (e, 2n) and (γ , 2n) reactions.

For the case of ^{181}Ta , the (γ , xn) cross-sections measured by Bergère et al. (1968) and the (e, xn) cross-sections measured by Cataldi et al. (1988) are plotted in Fig. 3. We evaluated the ratio between these cross-sections as a function of the photon or electron energy up to 20 MeV and results obtained are shown in Fig. 4. Overall, the contamination of the photoneutron fluxes by electroneutrons increases with the electron energy. More particularly, we can see that the emission of electroneutrons is roughly 1000 times less probable than the emission of photoneutrons at 10 MeV. This value turns out to be about 500 at 15 MeV and approximately 73 at 20 MeV. It is worth to emphasize that the value of 137 indicated in (NCRP, 1984) seems to only be appropriate for an energy close to 18 MeV. Moreover, as no (e, xn) cross-sections have ever been measured for any of the tungsten isotopes in the energy range of interest, we assume in first approach that the production of electroneutrons by tungsten is of the same order of magnitude as for ^{181}Ta .

Regarding the energy spectra and angular distributions of electroneutrons produced by electron accelerators in the 4–20 MeV range, little

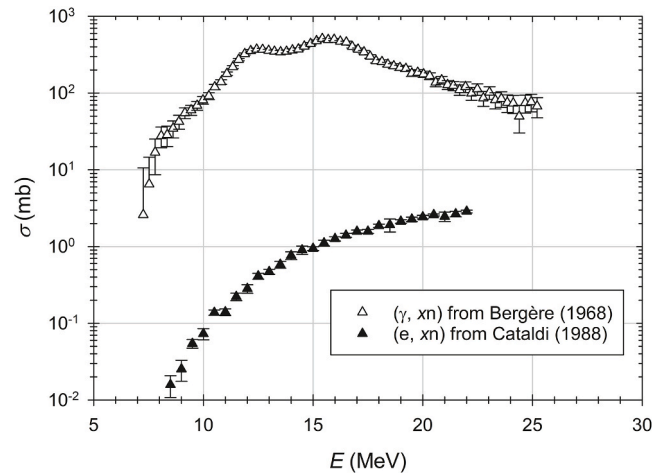


Fig. 3. Comparison of (γ , xn) cross-sections from Bergère et al. (1968) and (e, xn) cross-sections from Cataldi et al. (1988) for ^{181}Ta , as a function of the energy of the incident particle.

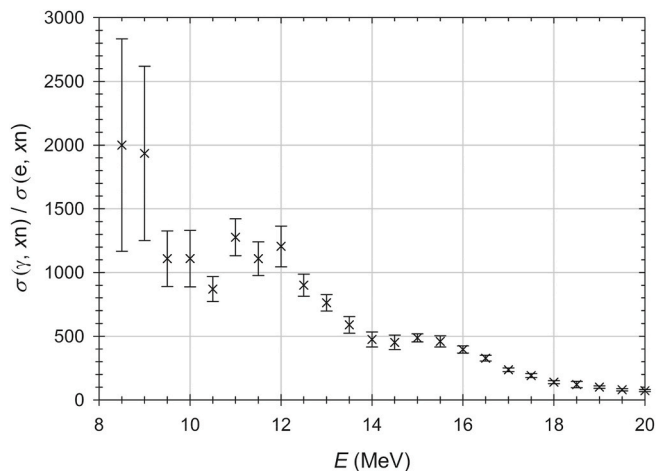


Fig. 4. Ratio between the (γ , xn) cross-sections from Bergère et al. (1968) and the (e, xn) cross-sections from Cataldi et al. (1988) for ^{181}Ta , as a function of the energy of the incident particle.

to no information exist. In the absence of published data, Kosako et al. (2011) assumed that the angular distribution of electroneutrons was the same as that of *Bremsstrahlung* photons, which means that they expected electroneutrons to be emitted preferably in the forward direction with respect to the electron beam. They also assumed that the energy of electroneutrons was the same as the maximum energy of photoneutrons, which they presumed equal to 7 MeV for a copper target bombarded by 18 MeV electrons. However, these expected energy spectrum and angular distribution have not been demonstrated and were selected without theoretical or empirical arguments. Therefore, on the basis of the *equivalent photon approximation* and considering the lack of experimental data, we recommend assuming in the meantime that electroneutron fluxes are characterized by the same energy spectra and angular distributions as photoneutron fluxes (see section 2.2).

When considering electron accelerators operating in the 4–20 MeV range and equipped with conversion targets, contamination of photoneutron fluxes by electroneutrons is expected to be limited. However, the emission of electroneutrons could become non-negligible when using electron accelerators in electron mode, i.e., without any conversion target, and potentially significant at high electron energies (≥ 30 MeV) at which the (e, xn) cross-sections may overtake the (γ , xn) cross-sections. Such a configuration without conversion target is notably encountered in the field of radiation therapy with medical accelerators operated in electron mode. The electroneutron production is foreseen to principally occur in the scattering foil, which is usually made of tungsten or tantalum and thick of a few tens of μm . In recent years, an increasing interest can be reported towards electroneutron production from medical accelerators (Vega-Carrillo and Pérez-Landeros, 2012; Biltekin et al., 2015; Soto-Bernal et al., 2017, 2018; Mathew et al., 2020). Vega-Carrillo and Pérez-Landeros (2012) measured the spectrum of electroneutrons in the vicinity of a 12 MeV medical accelerator (Varian 2100CD) using Bonner spheres. They reported that the spectrum was mostly composed of thermal and epithermal neutrons, which is in fact explained by the moderation of electroneutrons in the irradiation hall. The measurement of the initial energy spectrum of electroneutrons from a fundamental point of view or in other words at the time of their emission would require an adapted protocol and perhaps the use of a less complex accelerator geometry than the ones used in radiation therapy. In a broader scope of applications of electron accelerators, it is worth mentioning that Voronchev et al. (2005) claimed to reach an intensity of 2×10^{13} n/s thanks to the $^9\text{Be}(e, e'n)^8\text{Be}$ reaction by inducing a 10 MeV electron beam on a dense beryllium plasma. However, when coupling an electron accelerator operating in the 4–20 MeV range already equipped with a tungsten or tantalum target with a secondary target made of beryllium or heavy water, the production of electroneutrons is foreseen to be limited to the primary target. In other words, the production of electroneutrons in the beryllium or heavy water secondary target can be neglected.

In summary, the measurements of accurate fundamental data related to electroneutrons, such as: yield cross-sections, energy spectra and angular distributions; offer challenges that still have to be met as:

- it is not possible to detect and measure the energy of a virtual photon at the origin of the emission of an electroneutron;
- photoneutrons may also be produced along with electroneutrons, as electrons may undergo the *Bremsstrahlung* phenomenon and be responsible for the emission of photons that could in turn lead to the emission of photoneutrons;
- electroneutrons and photoneutrons are emitted simultaneously and could potentially have close (if not the same) characteristics in terms of energy spectra and angular distributions.

Finally, considering electron accelerators operating in the 4–20 MeV range equipped with electron-to-photon conversion targets, we estimate from Fig. 4 that contamination of photoneutron fluxes by electroneutrons is likely to be overall below 2% and sometimes as low as 0.05%

considering an accelerator operated at 9 MeV for instance.

2.4. Implementation in Monte Carlo codes

White et al. (1999) stated that photonuclear physics has been “the neglected stepchild of photon transport” in Monte Carlo simulation codes and blamed the lack of complete, evaluated photonuclear data. They present the early benchmarking tests conducted at the *Los Alamos National Laboratory* for the MCNP code and compare agreement obtained between MCNP calculations and experimental data published forty years earlier by Barber and George (1959), who measured neutron yields from thick targets of different materials bombarded by electrons in the 10–36 MeV range. Depending on the materials and the electron energies considered, the discrepancies between simulation results and experiments vary from a few percent to a few tens of percent at 19 MeV and reach large differences of the order of a factor of twenty at 10 MeV. White et al. (1999) reported that it was not clear whether the causes of these discrepancies were in the experiments, the evaluated data or the MCNP simulations.

In 2003, the same team published two companion papers concerning photonuclear physics in radiation transport. The first one (Chadwick et al., 2003) describes model calculations and nuclear data evaluations of photonuclear reactions on isotopes of C, O, Al, Si, Ca, Fe, Cu, Ta, W, and Pb for incident photon energies up to 150 MeV whereas the second one (White et al., 2003) describes the work done to process these files into transport libraries and to update the MCNP(X) code to use tabular photonuclear reaction data. White et al. (2003) also described initial validation simulations used to benchmark several of the photonuclear transport tables. Furthermore, Chadwick et al. (2003) pointed out that one of the main problems is that most photonuclear data have been measured with incident *Bremsstrahlung* photon spectra, and not with monoenergetic photons. This causes a problem because angular distributions are incident energy dependent as well as angle-energy correlated, i.e., the angular distribution varies for different secondary emission energies. This first paper also specifies that in the *Giant Dipole Resonance* regime, a simple approximation of isotropy is assumed except for carbon and oxygen for which experimental data are available and used. The second paper reports that the MCNP(X) codes can provide photoneutron production predictions with $\sim 25\%$ uncertainty. For the case of thin targets, White et al. (2003) reminded that electro-disintegration effects may be important and are ignored by the MCNP(X) code. The authors also stated that the calculations generally underestimate the experimental values, and claim to believe – although they admit that it would be extremely difficult to prove – that there is some systematic overestimation in the experimental values from Barber and George (1959). However, one would notice that the article does not mention what makes them believe in an overestimation trend in these measured values. Anyway, White et al. (2003) emphasize the need for the community to help provide thorough validation testing for photoneutron production capabilities and claim to hope that more benchmark experiments will be undertaken.

Heinrichs and Lent (2003) benchmarked the COG Monte Carlo code (Buck and Lent, 2002) and MCNPX (Pelowitz, 2011) against the experimental data from Barber and George (1959). In particular, they concluded that photoneutron yields calculated with COG and MCNPX are generally in good agreement with the measurements of Barber and George (1959) with no over-predicted yields. However, they warned the code users that calculated yields may be under-predicted by as much as 35% or by an order of magnitude at very low energies near the reaction threshold. In 2016, Frankl and Macián-Juan (2016) published an update – using the photonuclear cross-section library ENDF7U and the transport code MCNPX, version 2.7 – of the previous photonuclear benchmarks conducted by White et al. (2003). The paper also provides a first benchmark result with MCNPX for natural uranium. In line with White et al. (2003), Frankl and Macián-Juan (2016) concluded that a rather good agreement can be observed between their MCNPX calculations and

the measurements reported by Barber and George (1959) although a systematic underestimation in the calculated photoneutron yields can be stressed. Frankl and Macián-Juan (2016) admitted not knowing the reasons for this underestimation trend.

Regarding electroneutrons, Budnev et al. (1975) derived the *equivalent photon approximation* in expressions potentially applicable to the implementation of electronuclear reactions in Monte Carlo transport codes. At present, (e, n) or (e, e'n) electronuclear reactions are not implemented in most Monte Carlo codes, except for GEANT4 and FLUKA for instance (Tsechanski et al., 2020). Yet, we recommend using Monte Carlo simulation codes sparingly and with a high level of vigilance when aiming at simulating electroneutron fluxes produced by electron accelerators operating in the 4–20 MeV range. Indeed, fundamental knowledge on energy spectra and angular distributions, as well as validated electronuclear data and extensive Monte Carlo code benchmarking are all crucially lacking to judge as reliable simulation results about electroneutron production from electron accelerators.

Finally, regarding the physics of photoneutrons implemented in commonly used Monte Carlo simulation codes:

- for the angular distribution, isotropy is often assumed in the GDR regime for simplicity (IAEA, 2000);
- for the energy spectrum, different approaches are used. Information on the algorithms implemented in the codes are sometimes provided (Petit et al., 2011). However, when it is not the case, thorough investigations are necessary and may bring to light errors in commonly used approximations, as explained by Fynan (2020).

In other respects, the experimental data from Barber and George (1959) are not the only ones that exist to assess the reliability of Monte Carlo codes for the simulation of photoneutrons emitted by electron accelerators operating in the 4–20 MeV range. This point will be reviewed in detail in section 4 of this paper. But first, let us determine how one of the most known and widely used Monte Carlo codes – MCNP – simulates the photoneutron fluxes emitted by the four materials most commonly encountered in (or coupled with) electron accelerators, i.e., tungsten or tantalum and heavy water or beryllium.

3. Characterization of photoneutron fluxes by Monte Carlo simulation

3.1. Materials and methods

In section 3, we examine the characteristics of photoneutron fluxes emitted by electron accelerators in the 4–20 MeV range simulated with the MCNP Monte Carlo code, which is developed by the Los Alamos National Laboratory (New Mexico, USA) and widely used by the scientific community working with electron accelerators. When new and complementary calculations are presented, the latter have been carried out using the latest version of the code, i.e., MCNP6.2 (Werner, 2017, 2018) with nuclear data from the ENDF/B-VII.1 library (Chadwick et al., 2011), which contains evaluations based on nuclear models fitting multiple experimental data and sometimes for certain isotopes data from theoretical models in the absence of experimental data. In a first step, we characterize photoneutron fluxes emitted by tungsten or tantalum targets irradiated by electrons in the 7–20 MeV range. Fig. 5 shows a schematic representation of the corresponding MCNP simulation model. The target has a cylindrical geometry with a thickness of 5 mm and a diameter between 5 mm and 5 cm, and is irradiated orthogonally by monoenergetic electrons whose energy is between 7 and 20 MeV. In a second step, we characterize photoneutron fluxes emitted by heavy water or beryllium targets irradiated by monoenergetic photons in the 4–9 MeV range. A schematic representation of the corresponding MCNP simulation model is shown in Fig. 6. The target is of spherical geometry with a diameter fixed at 10 cm, and is irradiated by monoenergetic photons whose energy is between 4 and 9 MeV. In any case, the

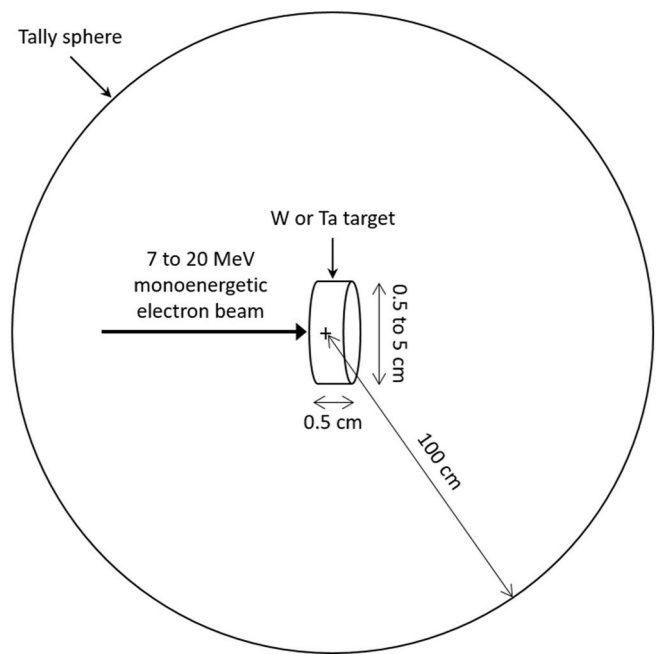


Fig. 5. Schematic representation of the MCNP simulation model of the tungsten or tantalum cylindrical target irradiated by monoenergetic electrons between 7 and 20 MeV (not to scale).

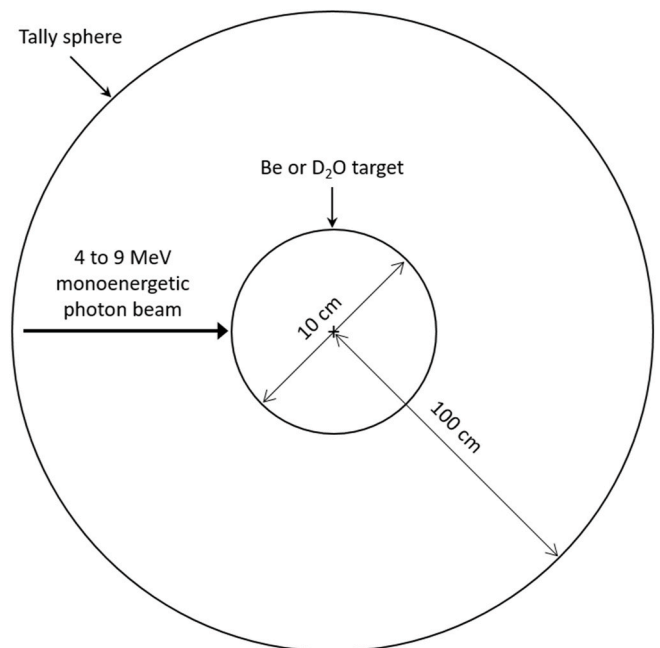


Fig. 6. Schematic representation of the MCNP simulation model of the beryllium or heavy water spherical target irradiated by monoenergetic photons between 4 and 9 MeV (not to scale).

photoneutron fluxes will be characterized according to three main parameters, i.e., photoneutron yield cross-sections, energy spectrum and angular distribution.

The MCNP code uses photoneutron yield cross-sections to simulate the production of photoneutrons. For verification purposes, we will extract the photoneutron yield cross-sections used by the code and compare them to cross-sections measured by different authors. It is important to highlight that according to this methodology the cross-sections extracted are the ones – precisely used by the code – and not the ones expected to be used and taken directly from the ENDF database. Let us describe the rate of a photonuclear reaction of interest

$$R = \int_0^{E_{\max}} \varphi(E) N \sigma(E) dE, \quad (9)$$

with:

- R , the photonuclear reaction rate of interest (in $\text{cm}^{-3} \cdot \text{s}^{-1}$);
- $\varphi(E)$, the photon flux (in $\text{cm}^{-2} \cdot \text{s}^{-1}$);
- N , the atomic density (in cm^{-3});
- $\sigma(E)$, the microscopic cross-section of the photonuclear reaction of interest (in cm^2).

The atomic density can be rewritten as

$$N = \frac{\rho N_A}{M}, \quad (10)$$

with:

- N , the atomic density (in cm^{-3});
- ρ , the material density (in $\text{g} \cdot \text{cm}^{-3}$);
- N_A , the Avogadro's constant ($\approx 6.022 \times 10^{23} \text{ mol}^{-1}$);
- M , the atomic mass (in $\text{g} \cdot \text{mol}^{-1}$).

Finally, the microscopic cross-section of a photonuclear reaction of interest for a photon of energy E can be extracted using the following formula

$$\sigma(E) = \frac{R}{\varphi(E) \times \frac{\rho N_A}{M}}. \quad (11)$$

In MCNP, $\varphi(E)$ can be calculated using an *F4 tally* in the volume of the target and R can be calculated using an *F4 tally* coupled with an *FM4 card*, in which is specified the reaction of interest via a reaction type number (MT) either defined according to the ENDF-6 format manual (Trkov and Brown, 2018) or the MCNP user's manual (Werner, 2017) for non-standard special reaction numbers. In the frame of this study, we will use:

- the sum of MT numbers from 50 to 91 to extract (γ, n) cross-sections (standard ENDF reaction numbers);
- the MT number 16 to extract $(\gamma, 2n)$ cross-sections (standard ENDF reaction number);
- the MT number 1001 to extract (γ, xn) cross-sections (non-standard special reaction number).

In order to determine the photoneutron energy spectrum and angular distribution, a sphere of 1-m radius is simulated with MCNP around the photoneutron-producing target (see Figs. 5 and 6). The photoneutron energy spectrum can be determined thanks to the number of photoneutrons crossing the sphere per source particle hitting the target, which will be calculated with an *F1 tally* and 10 keV energy bins indicated in an *E1 card*. The determination of the photoneutron angular distribution requests to set a 0° polar angle reference vector towards the source particle direction, which can be done using the *FRV option* of an *FT card*. The *F1 tally* can then be coupled with 4° polar angle bins described in a *C1 card*. Furthermore, considering an $[\alpha, \beta]$ angular bin, α being the lower angle and β the upper one, each number of photoneutrons crossing the sphere in an $[\alpha, \beta]$ angular bin will be divided by $2\pi[\cos(\alpha) - \cos(\beta)]$ in order to normalize the latter number by steradian.

3.2. Characteristics of photoneutron fluxes emitted by tungsten or tantalum targets

3.2.1. Photoneutron yield cross-sections

Errors were discovered and reported in (Sari et al., 2012) regarding the (γ, xn) cross-sections used by the MCNPX code, version 2.7.B (Pelowitz, 2011) especially around the energy threshold of the (γ, xn)

reactions for three of the five tungsten isotopes, i.e., ^{182}W , ^{184}W and ^{186}W . These errors lead to an underestimation of the photoneutron production by tungsten and more particularly at low energy. By way of example, this underestimation of the photoneutron production by a tungsten target irradiated by 10 MeV electrons exceeds a factor of three (Sari et al., 2012). Later on, these errors were identified in the frame of a collaboration with the development team of the TRIPOLI-4 Monte Carlo code (Sari et al., 2013b) and explained by the fact that some photoneutron yields (x) were set to zero instead of one at low energy in the nuclear data of the ENDF evaluation ($x = 0$ instead of $x = 1$). In this paper, we will investigate if these errors in the (γ, xn) cross-sections for tungsten isotopes have been corrected in the latest version of the MCNP code, i.e., MCNP6.2 (Werner, 2017, 2018).

However, before doing so, it is important to inform the reader that the photonuclear cross-sections from CNDC (Chinese Nuclear Data Centre) previously provided with MCNPX – in addition to the photoneutron data from ENDF – are no longer available with MCNP6.2. Furthermore, in order to circumvent the (γ, xn) cross-sections errors at low energy in MCNPX, a potential solution was provided in (Sari et al., 2013b) and consisted into calculating “manually” a (γ, xn) reaction rate thanks to the individual (γ, n) and $(\gamma, 2n)$ reactions rates using the cross-sections from CNDC. Thus, this solution working with MCNPX becomes inapplicable with MCNP6.2.

Figs. 7 and 8 present the (γ, n) , $(\gamma, 2n)$ and (γ, xn) photonuclear cross-sections extracted from MCNP6.2 using Eq. (11) respectively for ^{180}W

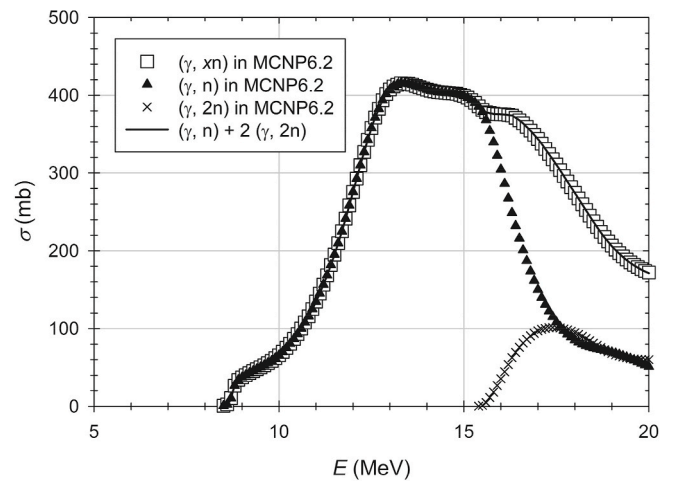


Fig. 7. Extraction of ^{180}W photonuclear cross-sections implemented in MCNP6.2, using Eq. (11).

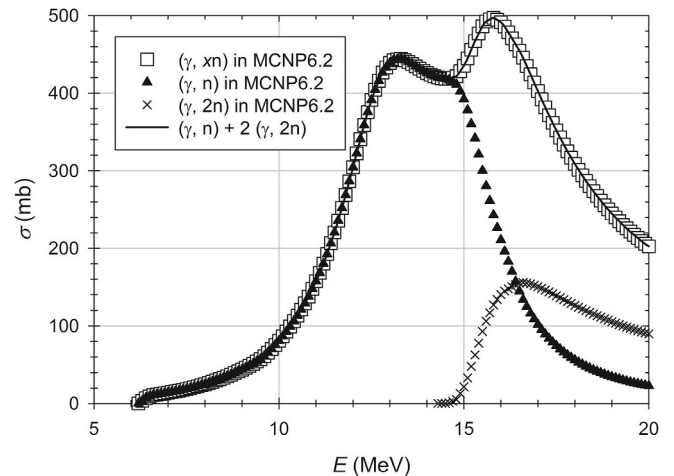


Fig. 8. Extraction of ^{183}W photonuclear cross-sections implemented in MCNP6.2, using Eq. (11).

and ^{183}W . The (γ, xn) cross-sections are also reconstructed “manually” by summing the (γ, n) and $(\gamma, 2n)$ cross-sections, including the neutron multiplicity. For both ^{180}W and ^{183}W , it is worth to notice that the (γ, xn) cross-sections are in excellent agreement with the $(\gamma, n) + 2 (\gamma, 2n)$ manual reconstruction. Nevertheless, it is important to highlight that, for these two isotopes, no experimental data are available (IAEA, 2000). Therefore, the ENDF evaluation has used a model to calculate and generate the photonuclear cross-sections for ^{180}W and ^{183}W .

The (γ, n) , $(\gamma, 2n)$ and (γ, xn) photonuclear cross-sections for ^{182}W extracted from MCNP6.2 using Eq. (11) are presented in Fig. 9. Two main comments can be made. First, the (γ, xn) cross-sections are in disagreement with the (γ, n) at low energy around the threshold. Second, the cross-sections measured by Goryachev and Zalesnyy (1978) enable to confirm that the erroneous cross-sections are the (γ, xn) ones. Fig. 10 presents the (γ, xn) photonuclear cross-sections for ^{184}W extracted from MCNP6.2 using Eq. (11). One would notice that the (γ, n) and $(\gamma, 2n)$ cross-sections are not available for this isotope in MCNP6.2. The cross-sections measured by Goryachev et al. (1973), by Veyssi re et al. (1975) and by Goryachev and Zalesnyy (1978) are in good agreement at low energy with the (γ, xn) cross-sections extracted from MCNP6.2. Nevertheless, the experimental data from Goryachev et al. (1973) suggest that the maximum of the (γ, xn) cross-sections curve around 15 MeV is undervalued, which could contribute to an underestimation of the photoneutron production by MCNP6.2 in the case of an electron

accelerator equipped with a tungsten target and operated above 14 MeV. While the first problem brought to light involved errors in the nuclear data of the MCNPX and MCNP6.2 codes, this second problem is different and relates to the lack of fundamental knowledge of photonuclear cross-sections. The (γ, n) , $(\gamma, 2n)$ and (γ, xn) photonuclear cross-sections for ^{186}W extracted from MCNP6.2 using Eq. (11) are presented in Fig. 11. The two comments previously made for ^{182}W are also relevant for ^{186}W . First, the (γ, xn) cross-sections are in disagreement with the (γ, n) at low energy around the threshold. Second, the cross-sections measured by Berman et al. (1969), by Goryachev et al. (1973) and by Goryachev and Zalesnyy (1978) enable to confirm once again that the erroneous cross-sections are the (γ, xn) ones. In other words, the (γ, xn) cross-section threshold errors for ^{182}W and ^{186}W discovered in MCNPX in 2012 (Sari et al., 2012) when using the ENDF library are unfortunately still relevant in MCNP6.2 in 2022. This conclusion illustrates the need for extensive investigations and validation steps in both the nuclear data libraries and their implementations in Monte Carlo codes.

In order to assess the impact of the (γ, xn) cross-section threshold errors for ^{182}W and ^{186}W on the photoneutron production in the case of a tungsten target simulated with MCNP6.2, first, we calculated the (γ, xn) reaction rate in a target made of natural tungsten, which is composed of the five tungsten isotopes with their relative abundancies. Second, we simulated individually five targets each one being composed at 100% of one of the isotope of tungsten. For the targets made of either ^{180}W , ^{183}W or ^{184}W , we calculated directly the (γ, xn) reaction rates in the targets, whereas for the targets made of either ^{182}W or ^{186}W , and in order to avoid the (γ, xn) cross-section threshold errors, we calculated “manually” the (γ, xn) reaction rate in the target by summing a (γ, n) and two $(\gamma, 2n)$ reaction rates. Third, the “corrected” (γ, xn) yield was obtained by summing according to their relative abundancies the “non-erroneous” (γ, xn) reaction rates obtained in the five individual targets. This comparison was conducted fourteen times for electron energies from 7 to 20 MeV by 1 MeV steps. The results obtained for a 5 mm-thick target are gathered in Table 2. At 7 MeV, the (γ, xn) and the “corrected” yields match perfectly as at this energy the only isotope contributing to the photoneutron production is ^{183}W . At 10 MeV, the underestimation of the photoneutron production due to the (γ, xn) threshold errors for ^{182}W and ^{186}W in MCNP6.2 reaches nearly a factor of two. Above 15 MeV, the underestimation is below 20%. It is important to highlight that the ratios provided in Table 2 may also be used by the reader as scaling factors in view of reevaluating photoneutron yields from tungsten targets irradiated by electrons in the 7–20 MeV range, which are underestimated in MCNP6.2.

In order to avoid the errors encountered with a tungsten target regarding the production of photoneutrons with the MCNP code, one

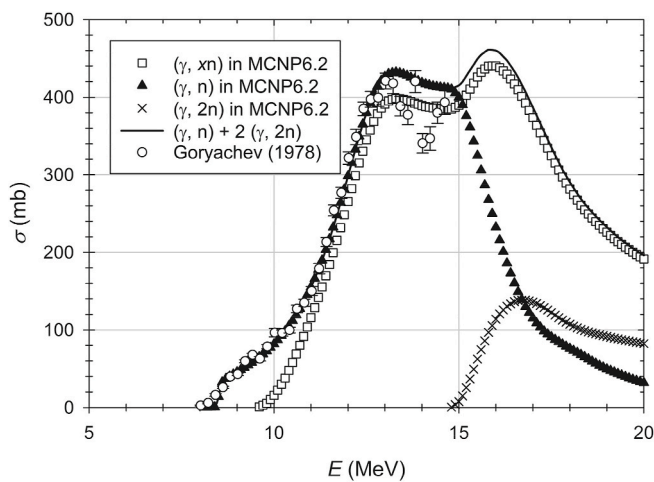


Fig. 9. Extraction of ^{182}W photonuclear cross-sections implemented in MCNP6.2, using Eq. (11), and comparison with experimental data.

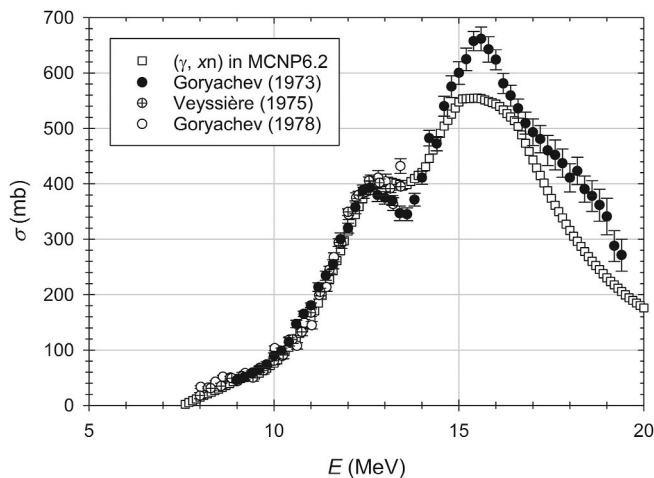


Fig. 10. Extraction of ^{184}W photonuclear cross-sections implemented in MCNP6.2, using Eq. (11), and comparison with experimental data.

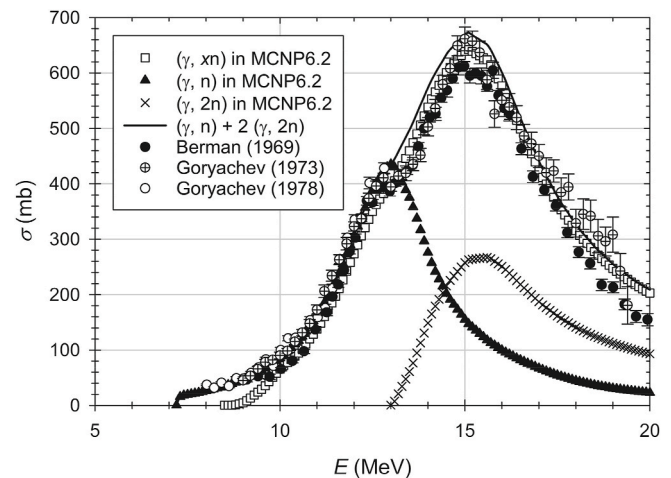


Fig. 11. Extraction of ^{186}W photonuclear cross-sections implemented in MCNP6.2, using Eq. (11), and comparison with experimental data.

Table 2

Photoneutron yields emitted by a tungsten target irradiated by monoenergetic electrons in the 7–20 MeV range: ratios between the photoneutron yields calculated with MCNP6.2's erroneous (γ, xn) cross-sections and calculated according to the methodology proposed in this paper to avoid the (γ, xn) cross-section threshold errors.

Electron energy (MeV)	(γ, xn) yield corrected yield
7	1.00
8	0.69
9	0.56
10	0.53
11	0.58
12	0.66
13	0.74
14	0.75
15	0.83
16	0.86
17	0.87
18	0.89
19	0.90
20	0.90

could be tempted to use a tantalum target instead. Fig. 12 presents the (γ, xn) photonuclear cross-sections for ^{181}Ta extracted from MCNP6.2 using Eq. (11). One would also notice that the (γ, n) and $(\gamma, 2n)$ cross-sections are not available for this isotope in MCNP6.2. The cross-sections measured by Bergère et al. (1968) are roughly in agreement with the (γ, xn) cross-sections extracted from MCNP6.2 at both low energy and around the maximum of the (γ, xn) cross-section curve around 16 MeV. Nevertheless, few experimental data exist concerning the photonuclear cross-sections for ^{181}Ta and the use of a tungsten conversion target is the most common industrial solution chosen by electron accelerator manufacturers.

3.2.2. Energy spectrum

We simulated with MCNP6.2 the energy spectrum of the photoneutron fluxes emitted by both tungsten and tantalum targets irradiated by a 15 MeV monoenergetic electron beam. Fig. 13 presents the results obtained. As discussed in section 2.2, the photoneutron energy spectrum is foreseen to follow a Maxwell-Boltzmann curve. Nevertheless, the shape of the energy spectrum obtained for the case of a tantalum target below 0.5 MeV is not typical of such a Maxwell-Boltzmann curve, which was already observed with the MCNPX code in 2013 (Sari et al., 2013b). Fig. 14 shows the energy spectrum of the photoneutron fluxes emitted by a tantalum target irradiated by 9–18 MeV electrons by 3 MeV steps,

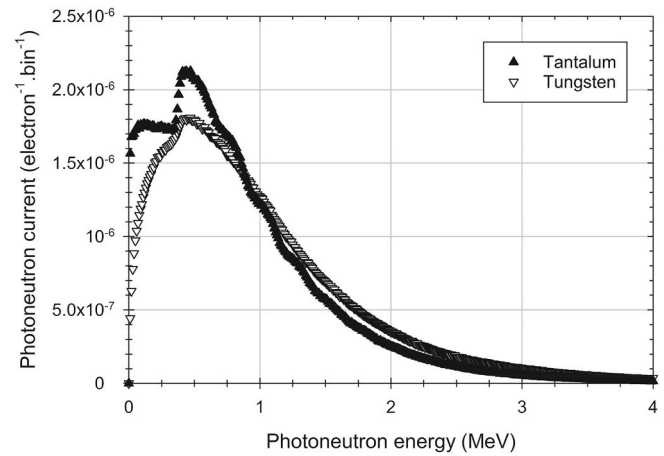


Fig. 13. Energy spectrum of photoneutrons emitted by the tungsten or tantalum target irradiated by 15 MeV electrons, simulated with MCNP6.2.

simulated with MCNP6.2. The strange shape of the energy spectrum obtained for a tantalum target under 0.5 MeV appears to be present whatever the electron energy. It is likely that this shape does not have any physical interpretation, and could be due to inaccurate algorithms dedicated to the photoneutron production in the MCNP code or errors in the nuclear data.

3.2.3. Angular distribution

We also simulated with MCNP6.2 the angular distribution of the photoneutron fluxes emitted by both tungsten and tantalum targets irradiated by 15 MeV monoenergetic electrons. Fig. 15 presents the results obtained for the case of a 5 cm-diameter target. Two comments can be made. First, the two energy spectra overlap each other. Second, a gap appears at 90°, along the diameter of the target. Fig. 16 shows the angular distribution of the photoneutron fluxes emitted by tungsten targets of either 5 mm or 5 cm of diameter irradiated by 15 MeV monoenergetic electrons, simulated with MCNP6.2. The angular distribution obtained for a target of 5 mm of diameter is isotropic, thereby proving that the gap observed at 90° for the case of a target of 5 cm of diameter is due to target geometry effects. As discussed in section 2.2, knock-on photoneutrons emitted by a direct emission process are expected to be present in the photoneutron energy spectrum above a few MeV and characterized by an anisotropic angular distribution, with a privileged emission angle around 90° from the incident photon. Fig. 17 presents the normalized angular distribution of the photoneutron flux components

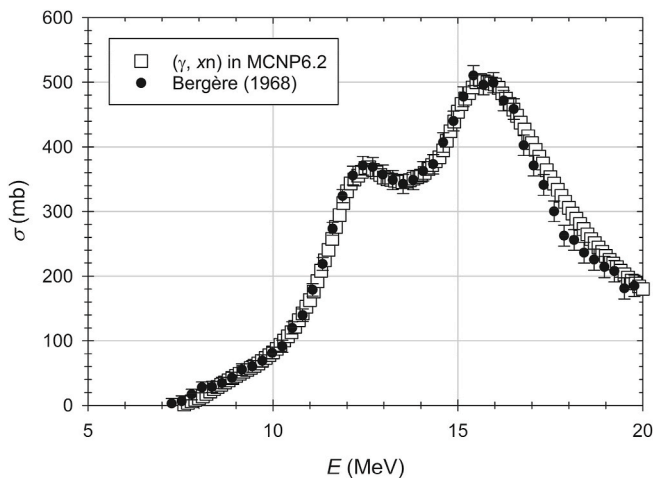


Fig. 12. Extraction of ^{181}Ta photonuclear cross-sections implemented in MCNP6.2, using Eq. (11), and comparison with experimental data.

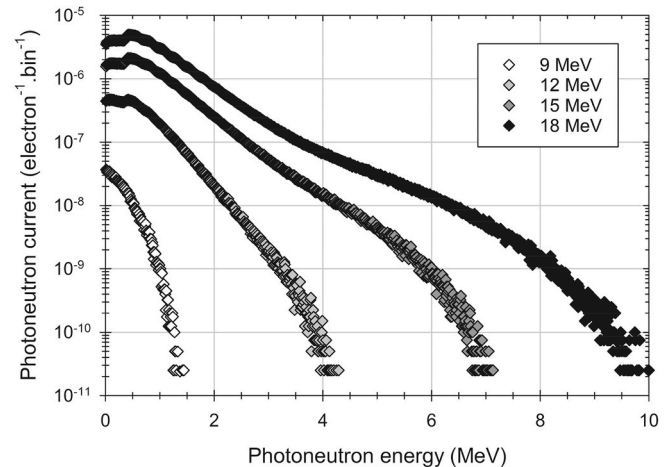


Fig. 14. Energy spectrum of photoneutrons emitted by the tantalum target irradiated by either 9, 12, 15 or 18 MeV electrons, simulated with MCNP6.2.

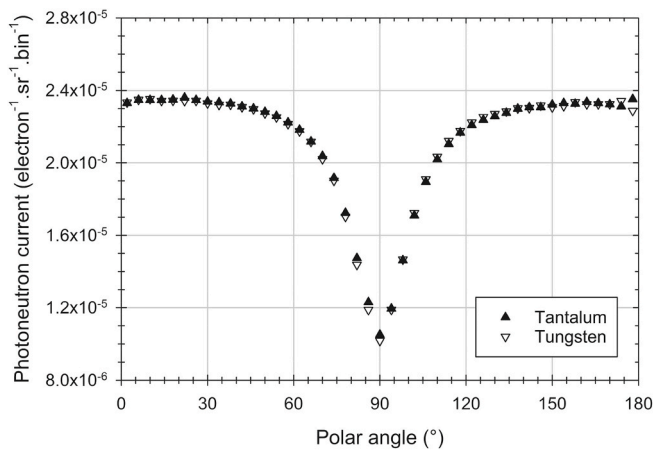


Fig. 15. Angular distribution of photoneutrons emitted by the tantalum and tungsten targets of 5 mm of thickness and 5 cm of diameter irradiated by 15 MeV monoenergetic electrons, simulated with MCNP6.2.

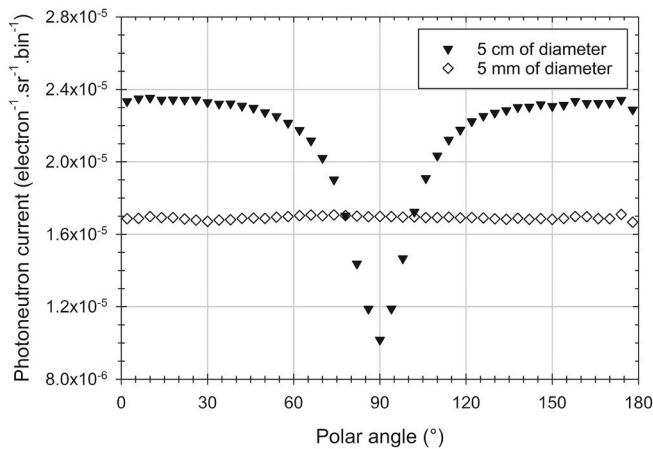


Fig. 16. Angular distribution of photoneutrons emitted by the tungsten targets of 5 mm of thickness and either 5 mm or 5 cm of diameter irradiated by 15 MeV monoenergetic electrons, simulated with MCNP6.2.

from 0 to 5 MeV and from 5 to 20 MeV, emitted by a 5 mm-diameter tungsten target irradiated by 15 MeV monoenergetic electrons, simulated with MCNP6.2. The 5–20 MeV photoneutron component appears

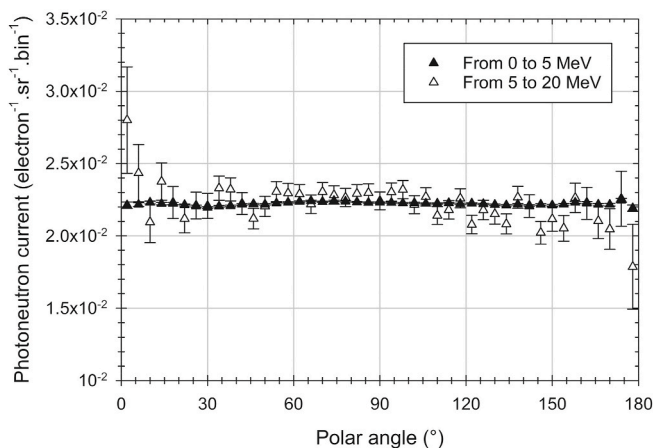


Fig. 17. Normalized angular distribution of photoneutrons with energies from 0 to 5 MeV and from 5 to 20 MeV, emitted by the 5 mm-diameter tungsten target irradiated by 15 MeV monoenergetic electrons, simulated with MCNP6.2.

to be isotropic as well as the 0–5 MeV component. We can infer from these results that the MCNP code does not simulate the direct component of photoneutron fluxes with its specific angular distribution.

3.3. Characteristics of photoneutron fluxes emitted by heavy water or beryllium secondary targets

3.3.1. Photoneutron yield cross-sections

As a reminder from section 1, a heavy water secondary target enables to produce photoneutrons through (γ, n) reactions on deuterium nuclei contained in the latter. Obviously, in the case of a heavy water secondary target, $(\gamma, 2n)$ reactions are prohibited as deuterium (^2H) is composed of a single neutron. It should also be stressed that in the 4–9 MeV range, production of photoneutrons from ^{16}O (abundance of 99.76%) contained in heavy water is also prohibited due to a (γ, n) reaction energy threshold set at 15.66 MeV (IAEA, 2000). For simplicity, other oxygen isotopes are not simulated in the following MCNP calculations. The (γ, n) photonuclear cross-sections for ^2H extracted from MCNP6.2 using Eq. (11) are presented in Fig. 18. Experimental data measured by Ahrens et al. (1974), by Birenbaum et al. (1985), by Hara et al. (2003) and the data from the IAEA/PD-2019 library (Kawano et al., 2020) are also plotted on this graph. We can deduce from this figure that the maximum of the (γ, n) cross-section curve around 4.5 MeV is undervalued. In other words, the intensity of the photoneutron flux emitted by a heavy water target coupled with an electron accelerator operating in the 4–9 MeV range is likely to be underestimated when simulated with the MCNP code.

The (γ, xn) photonuclear cross-sections for ^9Be extracted from MCNP6.2 using Eq. (11) are presented in Fig. 19. Experimental data measured by Kneissl et al. (1975), by Utsunomiya et al. (2015), and the data from the JENDL/PD-2004 library (Kishida et al., 2005) are also plotted on this graph. This figure brings to light significant disagreements between the (γ, xn) cross-sections implemented in MCNP6.2 and especially with both the measurements from Utsunomiya et al. (2015) and the data from the JENDL library. We therefore recommend conducting MCNP simulations involving a beryllium secondary target coupled with an electron accelerator operating in the 4–9 MeV range with the utmost attention given the dubious reliability of the (γ, xn) cross-sections for ^9Be implemented in MCNP6.2.

3.3.2. Energy spectrum

We simulated with MCNP6.2 the energy spectrum of the photoneutron fluxes emitted by both beryllium and heavy water secondary targets irradiated by a 9 MeV monoenergetic photon beam. Fig. 20 presents the results obtained. The mean energy of the two spectra are

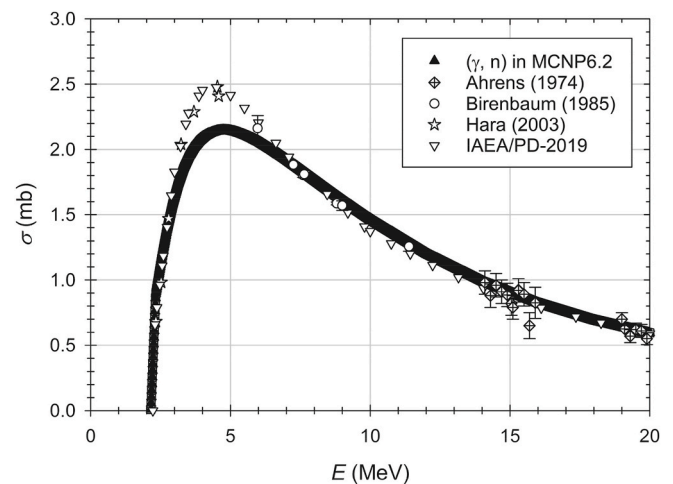


Fig. 18. Extraction of ^2H photonuclear cross-sections implemented in MCNP6.2, using Eq. (11), and comparison with experimental data.

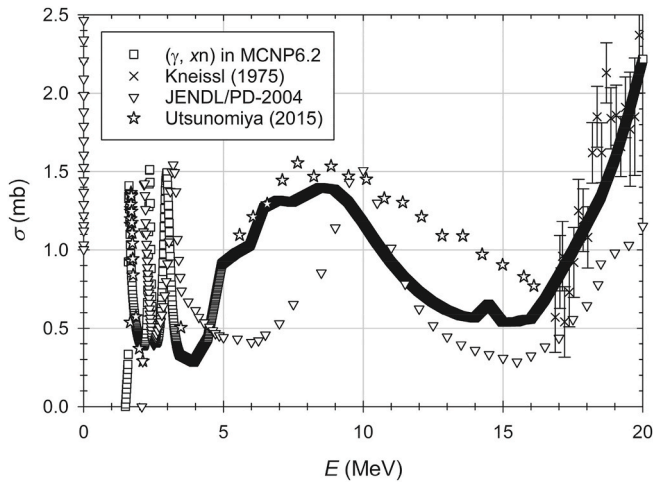


Fig. 19. Extraction of ^9Be photonuclear cross-sections implemented in MCNP6.2, using Eq. (11), and comparison with experimental data.

respectively 2.6 and 3.5 MeV for the beryllium and heavy water targets. Fig. 21 shows the energy spectrum of the photoneutron fluxes emitted by a heavy water secondary target irradiated by monoenergetic photons between 4 and 9 MeV by 1 MeV steps. The energy spectrum appears to be significantly harder when the photon energy increases. One would notice a certain number of dips in the energy spectra, for instance at 0.4, 1.0, 1.3, 1.65, 1.85 and 1.9 MeV. The latter are due to elastic scattering reactions of neutrons on ^{16}O contained in the heavy water target. To a lesser extent, (n, γ) reactions on ^{16}O may also occur and contribute marginally to the dips observed at around 0.4, 1.0, 1.65 and 1.85 MeV (Sari et al., 2020b).

3.3.3. Angular distribution

We also simulated with MCNP6.2 the angular distribution of the photoneutron fluxes emitted by secondary targets irradiated by monoenergetic photons between 4 and 9 MeV by 1 MeV steps. Fig. 22 presents the results obtained for a secondary target made of heavy water whereas Fig. 23 shows the results obtained for one made of beryllium,

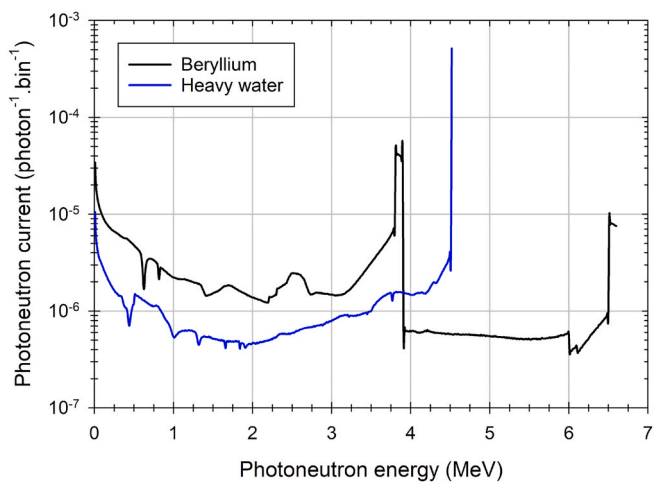


Fig. 20. Energy spectrum of photoneutrons emitted by the heavy water and beryllium secondary targets irradiated by 9 MeV monoenergetic photons, simulated with MCNP6.2.

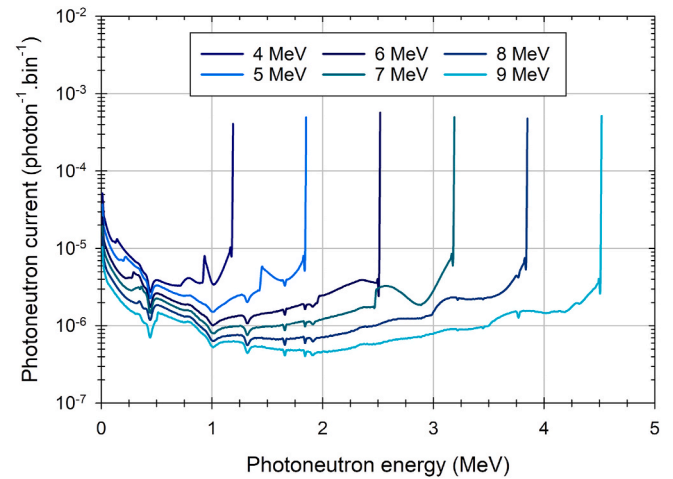


Fig. 21. Energy spectrum of photoneutrons emitted by the heavy water secondary target irradiated by monoenergetic photons between 4 and 9 MeV by 1 MeV steps, simulated with MCNP6.2.

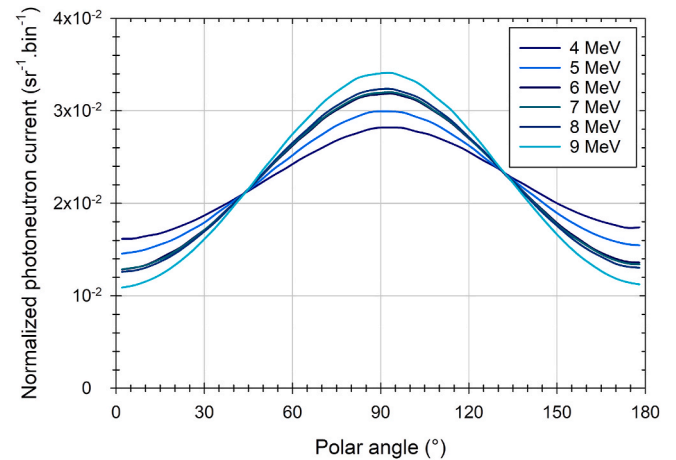


Fig. 22. Normalized angular distribution of photoneutrons emitted by the heavy water secondary target irradiated by monoenergetic photons between 4 and 9 MeV by 1 MeV steps, simulated with MCNP6.2.

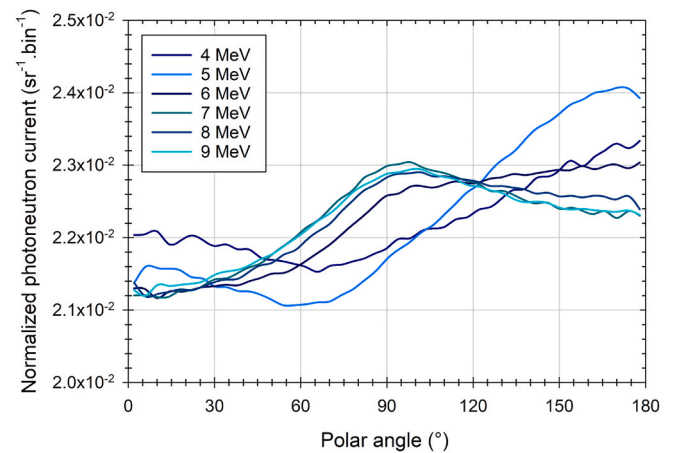


Fig. 23. Normalized angular distribution of photoneutrons emitted by the beryllium secondary target irradiated by monoenergetic photons between 4 and 9 MeV by 1 MeV steps, simulated with MCNP6.2.

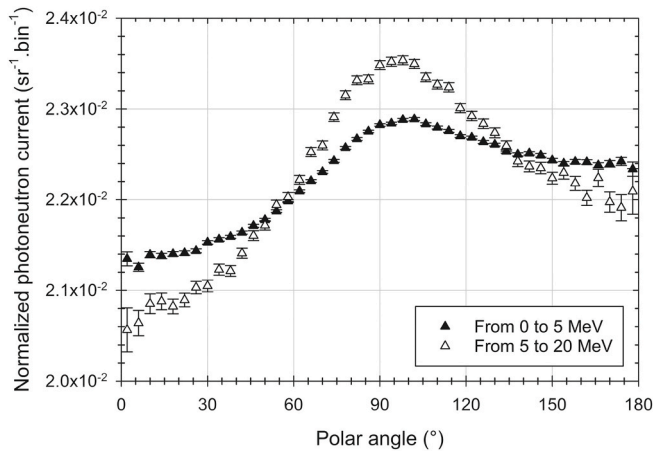


Fig. 24. Normalized angular distribution of photoneutrons with energies from 0 to 5 MeV and from 5 to 20 MeV, emitted by the beryllium secondary target irradiated by 9 MeV monoenergetic photons, simulated with MCNP6.2.

normalized distributions in all cases. The angular distribution of the photoneutron flux emitted by a heavy water secondary target appears to be peaked at 90° and the higher the photon energy the more this effect is pronounced (as a reminder, all the photoneutron emission is due to deuterium in the simulation cases with heavy water and contribution from ^{16}O is not possible). However, for the case of a beryllium secondary target, this effect is less marked but still noticeable for photon energies from 6 to 9 MeV. Fig. 24 presents the angular distribution of the components of the photoneutron flux from 0 to 5 MeV and from 5 to 20 MeV, emitted by a beryllium secondary target irradiated by 9 MeV monoenergetic photons. Both energy components appear to be peaked around 90°. Nevertheless, the most energetic component from 5 to 20 MeV is more strongly peaked at 90°.

4. Monte Carlo simulation of photoneutron fluxes: benchmarking MCNP against neutron activation measurements

4.1. Materials and methods

In section 4, we will review the performance of the MCNP code to simulate photoneutron fluxes emitted by electron accelerators in the 4–20 MeV range. To evaluate the reliability of the characteristics of the photoneutron fluxes simulated with MCNP, the latter can be benchmarked against neutron activation measurements of vanadium and aluminium foils (Sari et al., 2013c, 2018). The principle of the measurement is as follows. First, a foil is exposed to the photoneutron flux to measure its yield, leading to the activation of the foil, the production of measurable radioactive nuclei, which decay emitting gamma-rays of suitable energy and according to an appropriate half-life time. Then, the foil is positioned in front of a detector to acquire a gamma-ray spectrum and analyse the net peak area corresponding to the gamma of interest. To allow the comparison between the net peak areas simulated with the ones obtained experimentally, the net peak area is calculated with the following formula (Sari et al., 2013c, 2018, 2020b) that is valid for short-pulse irradiations

$$\mathcal{A}_\gamma = R_{(n,\gamma)} I_\gamma \varepsilon_\gamma \frac{1 - e^{-n\lambda T}}{1 - e^{-\lambda T}} t_1 e^{-\lambda t_2} (1 - e^{-\lambda t_3}), \quad (12)$$

with:

- $\mathcal{A}_\gamma$, the net peak area;
- $R_{(n,\gamma)}$, the (n,γ) reaction rate in the vanadium or aluminium foil;
- I_γ , the emission intensity of the gamma-ray of interest;
- ε_γ , the detection efficiency of the gamma detector;
- n , the number of pulses during the irradiation;
- λ , the decay constant of the radioactive nucleus;
- T , the pulse period;
- t_1 , the pulse duration;
- t_2 , the cooling time;
- t_3 , the counting time.

In this formula, the (n,γ) reaction rate in the foil is calculated with the MCNP code using an *F4* tally for incident neutrons in the volume of the foil coupled with an *FM4* card that specifies the reaction and material of interest. Other terms of the equation either depend on the experimental protocol or are extracted from nuclear databases (Sari et al., 2013c, 2018, 2020b).

4.2. Results

4.2.1. Photoneutron flux emitted by a tungsten or tantalum target

The benchmarking methodology presented in section 4.1 has been deployed on a Sagittaire accelerator (CEA Paris-Saclay, France) equipped with either a tungsten or tantalum target and operated at average electron energies of 14 or 17 MeV (Sari et al., 2013c). The vanadium or aluminium foils were positioned in a polyethylene box and surrounded by cadmium, see Figs. 1 and 2 from (Sari et al., 2013c), the whole constituting an activation detector. Table 3 gathers the results obtained using cross-sections for the (n,γ) reactions on ^{51}V and ^{27}Al from the ENDF/B-VI nuclear data library. The relative deviations between results obtained using (n,γ) cross-sections from ENDF/B-VI and ENDF-92 are within 8 and 18% (Sari et al., 2013c). Whatever the average electron energy, results obtained with the tungsten and the tantalum targets show similar trends. Indeed, calculations carried out took into account two energy-dependent correction factors to compensate the erroneous thresholds of the (γ, xn) cross-sections encountered with a tungsten target when using the MCNP code. At 14 MeV, the MCNP code underestimates experiment by roughly 40%. At 17 MeV, this underestimation reaches a factor of two. Moreover, additional measurements carried out with the Linatron M9 accelerator of the SAPHIR platform (CEA Paris-Saclay, France) equipped with a tungsten target and collimator (without any secondary target) and operated at an electron energy of 9 MeV suggest that for this material and at this energy the underestimation rises to a factor of the order of ten (Sari et al., 2018).

4.2.2. Photoneutron flux emitted by a heavy water secondary target

The benchmarking methodology has also been deployed on the Linatron M9 accelerator of the SAPHIR platform (CEA Paris-Saclay, France) operated at electron energies of 6 or 9 MeV and equipped with a 16 kg-heavy water secondary target (Sari et al., 2018). The vanadium or aluminium foils were positioned inside a neutron cell made of polyethylene, the design of which was detailed in (Sari et al., 2014), see Figs. 7 and 8 from (Sari et al., 2018). Table 4 summarizes the results obtained. At 6 MeV, the MCNP code underestimates experiment by a factor of two. At 9 MeV, this underestimation is around 26%.

4.2.3. Photoneutron flux emitted by a medical accelerator

As reported in (Sari et al., 2013c), this benchmarking methodology has also been deployed on a Saturne 43F medical accelerator designed by General Electric (CEA Paris-Saclay, France) operated in photon mode at 12 or 20 MV accelerating voltages. Calculations were carried out using cross-sections for the (n,γ) reactions on ^{51}V and ^{27}Al from the

Table 3

Ratio between simulation and experimental results, the Sagittaire accelerator of the SAPHIR platform (CEA Paris-Saclay, France) equipped with either a tungsten or tantalum target, and operated at average electron energies of 14 or 17 MeV (Sari et al., 2013c).

	14 MeV		17 MeV	
	Tungsten	Tantalum	Tungsten	Tantalum
V	0.56	0.56	0.52	0.51
Al	0.65	0.59	0.49	0.47
Mean value	0.61	0.58	0.51	0.49

Table 4

Ratio between simulation and experimental results, the Linatron M9 accelerator of the SAPHIR platform (CEA Paris-Saclay, France) equipped with a heavy water secondary target, and operated at electron energies of 6 or 9 MeV (Sari et al., 2018).

	6 MeV	9 MeV
V	0.48	0.73
Al	0.51	0.75
Mean value	0.50	0.74

Table 5

Ratio between simulation and experimental results, a Saturne 43F medical accelerator (CEA Paris-Saclay, France) operated in photon mode at either 12 or 20 MV accelerating voltages and vanadium or aluminium foils irradiated in the photon field (Sari et al., 2013c).

	12 MV	20 MV
V	0.12	0.79
Al	–	0.83
Mean value	–	0.81

ENDL-92 nuclear data library, which was released by the Lawrence Livermore National Laboratory (LLNL, USA). Table 5 gathers the results obtained with the activation detector irradiated in the photon field while containing either a vanadium or an aluminium foil. Due to low counting statistics, measurements conducted with an aluminium foil at 12 MV were not exploitable. At 12 MV, the MCNP code underestimates experiment by a factor of the order of ten. At 20 MV, this underestimation is around 20%.

4.3. Discussion

Whatever the energy at which electron accelerators are operating in the 4–20 MeV range, photoneutron activation calculations based on Monte Carlo simulations conducted with the MCNP code underestimate the experimental results. At first sight, the yields of the photoneutron fluxes simulated with MCNP seem to be underestimated, assumption that we will further investigate in the following. Considering the Linatron M9 accelerator operated at 9 MeV, photoneutrons are produced by tungsten and the underestimation climbs to a decade. Obviously, the (γ , xn) cross-section threshold errors met with the MCNP code (see section 3.2.1) should explain at least part of this discrepancy. Nevertheless, when tungsten is the sole photoneutron-emitting material in the setup simulated, a correction factor may be used *a posteriori* to circumvent the (γ , xn) cross-section threshold errors in MCNP. However, even if such a correction factor is taken into account when simulating a tungsten target, MCNP simulation results keep underestimating the experimental data by a few tens of percent in the 10–20 MeV range and the discrepancies are the same that the target is made of tungsten or tantalum (see Table 3). Thus, one could consider comparing results obtained with another Monte Carlo transport code.

A benchmark study between the Monte Carlo codes TRIPOLI-4 version 9 (Petit et al., 2011) and MCNPX version 2.7.B (Pelowitz, 2011) for the photoneutron yields of tungsten and tantalum targets

irradiated by electrons in the 10–20 MeV range is reported in (Sari et al., 2013b). This study showed that the gaps between results obtained with these two codes is overall around 10%, the yields determined with TRIPOLI-4 being lower than the ones obtained with MCNP. Consequently, it is likely that the underestimation obtained in the frame of photoneutron activation calculations carried out with MCNP would become even larger using TRIPOLI-4. On the other hand, Kim et al. (2016) simulated the photoneutron yields from tungsten, tantalum, lead, rhenium and gold targets irradiated by 10 or 15 MeV electrons and compared results obtained with two codes, i.e., MCNPX (Pelowitz, 2011) and GEANT4 (Agostinelli et al., 2003; Allison et al., 2006, 2016), see in particular Table 2 from (Kim et al., 2016). The discrepancies encountered between the two codes are diverse and uneven, starting as low as 6% and up to a factor of nearly three. The explanation put forward by Kim et al. (2016) to interpret these discrepancies is the lack of comprehensive understanding in physics of the photonuclear process and secondary photoneutrons.

In any case, the underestimation trend encountered with the MCNP code when simulating photoneutron activation measurements could be at least partially due to either or both of the following causes:

- lack of fundamental knowledge of photonuclear cross-sections near the energy thresholds of the reactions;
- undervaluation of the maximum of the photonuclear cross-section curves (as previously and suspiciously shown for ^{184}W in Fig. 10 for instance).

By way of illustration regarding the first hypothesis, a difference exceeding 0.5 MeV was pointed out by Lee et al. (1998) at the energy threshold of the (γ , n) reaction for ^{181}Ta when comparing experimental data measured at Saclay (7.45 ± 0.2 MeV) and Livermore (8.0 MeV). As a reminder, at 14 and 17 MeV, the underestimations met with the MCNP code when simulating neutron activation measurements are respectively around 40% and approximately at a factor of two for both tungsten and tantalum (see Table 3). Three important comments can be made. First, it is remarkable that the underestimation is higher at 17 than 14 MeV, which at least supports the second assumption. Second, if tungsten and tantalum photonuclear cross-sections are undervalued in the energy range of interest, the order of magnitude of this undervaluation seems to be rather similar for the two materials. Third, the underestimations obtained at 14 and 17 MeV initially reported in (Sari et al., 2013c) are in good agreement with results obtained by Kosako et al. (2011) who compared neutron activation calculations based on MCNP5 simulations and experiments using an electron accelerator operated at 18 MeV. Indeed, Kosako et al. (2011) showed that the calculations underestimate the measured data by a factor of 1.5–2.5 at 18 MeV. It is worth to notice that given these discrepancies, Kosako et al. (2011) concluded that "the reevaluations of the photoneutron production cross-sections of copper and tungsten isotopes below 20 MeV are necessary in view of improving the calculation accuracy".

Regarding the photoneutron fluxes generated by medical accelerators, the photonuclear reactions mainly occur in the jaws, the primary collimator and the target. Due to the complex geometries of such accelerators, different materials like tungsten, lead, steel, etc., are often involved in the production of photoneutrons. For instance, the Saturne 43F medical accelerator emits photoneutrons through photonuclear reactions on both tungsten and lead, and the jaws of the accelerator emit about half of the photoneutron flux, see Table IV and Fig. 10 from (Sari et al., 2013c). Such diversity in terms of materials and neutron emitting elements renders non-trivial the application of a correction factor on simulation results during the post-processing step to compensate the (γ , xn) cross-section threshold errors for tungsten in MCNP. Considering the Saturne 43F accelerator operated at 12 MV, the underestimation of the order of a factor of ten (see Table 5) could be explained at least by the cumulative effects of both (γ , xn) cross-section threshold errors for tungsten in the nuclear data from the MCNP code and questionable

evaluations of (γ , xn) cross-sections for tungsten and lead.

When it comes to photoneutrons emitted by a heavy water secondary target, simulation of neutron activation measurements carried out using MCNP6.1 (Goorley et al., 2012) underestimates experiment by a factor of two at 6 MeV and by 26% at 9 MeV (see Table 4). These discrepancies seem to show that the higher the electron energy, the more reliable is the photoneutron yield from the heavy water target simulated with the MCNP code. One could be tempted to use these values as scaling factors to reevaluate the photoneutron yields simulated with MCNP in a post-processing step. However, this approach would only be correct if the discrepancies found with MCNP are only and exclusively due to underestimations of the photoneutron yields, i.e., without any impact from the photoneutron energy spectrum on these discrepancies. Furthermore, a recent feasibility study in view of conducting linac-based neutron activation analysis to detect and quantify copper in scrap metal involved a 16 kg heavy water secondary target coupled with the Linatron M9 accelerator of the SAPHIR platform (CEA Paris-Saclay, France) operated at 6 MeV (Sari et al., 2020b). Using MCNP6.2 (Werner, 2017, 2018), this study showed that simulation of neutron activation measurements of copper underestimate experiment roughly by a factor of two, which corroborates prior conclusions provided in (Sari et al., 2018) and according to which the (γ , n) reaction cross-sections for deuterium are likely to be undervalued in the ENDF library. It is worth to recall that in section 3.3.1, Fig. 18 illustrated a noticeable undervaluation of the (γ , n) reaction cross-sections for deuterium provided in ENDF. Nevertheless, algorithms used to simulate photoneutron production should also be carefully reexamined as some erroneous equations for photoneutron production can be found implemented in Monte-Carlo simulation codes, even in widely and commonly used ones such as MCNP (Fynan, 2020).

Fynan (2020) reported that numerous works (Mueck and Bensch, 1973; Reeder and Warner, 1981; Gallmeier, 1995; Watanabe et al., 2009; Patil et al., 2010; Rahmani and Shahriari, 2010; Tatari and Ranjbar, 2014; Vega-Carrillo and Martinez-Ovalle, 2016; Rafiei and Tavakoli-Anbaran, 2018) introduce the Wattenberg or Hanson approximations to describe photoneutron kinematics but without discussion nor derivation. A few years earlier, Caro (2016) had found that the photoneutron energy-angle relationship was modelled in MCNP5 (X-5 Monte Carlo Team, 2003) using *neutron inelastic scattering* equations, which leads to significant errors especially when the target nucleus is a light particle, e.g., deuterium. Regarding the MCNP code, whatever the version considered among which MCNP5 (X-5 Monte Carlo Team, 2003), MCNPX (Pelowitz, 2011), MCNP6.1 (Goorley et al., 2012), MCNP6.2 (Werner, 2017, 2018), photoneutron kinematics modelling is not documented. We suspect that the same algorithms are used to model photonuclear physics in these different versions of the MCNP code, even though slight differences can be found regarding the photonuclear data available. For instance, photonuclear cross-sections from CNDC are available in MCNPX but not in MCNP6. Anyway, the investigations carried out by Caro (2016) and Fynan (2020) enabled to identify with a high level of confidence the equations erroneously implemented in the MCNP code to model photoneutron kinematics, see Eqs. (30) and (32) from (Fynan, 2020). Inelastic neutron scattering equations are inappropriate to describe photoneutron kinematics, which leads to wrong and significantly hardened photoneutron energy spectra. Fynan (2020) suspects that discrepancies between simulation and experiment reported in (Sari et al., 2018) are dominated by spectral effects in the simulation data induced by the incorrect photoneutron kinematics modeling in the MCNP6 code. Nevertheless, inaccuracies of the photonuclear reaction cross-sections may also somehow contribute to the discrepancies encountered between MCNP simulation and experiment when dealing with photoneutrons emitted by an electron accelerator operating in the 4–20 MeV range, in the context in which the reliability of photonuclear data has been openly discussed by Varlamov (2019).

5. General discussion

We examined the performance of Monte Carlo transport codes to simulate photoneutron fluxes emitted by electron accelerators operating between 4 and 20 MeV. In this energy range and from a theoretical point of view, photoneutrons are created through the *giant dipole resonance* regime of the total photonuclear cross-section. These photoneutrons can be emitted by two different processes, each giving them specific properties:

- “boiled off photoneutrons” are emitted by *evaporation process*, and are characterized by lower energies and an isotropic emission;
- “knock-on photoneutrons” are emitted by *direct emission process*, and are characterized by higher energies and an anisotropic emission (in $\sin^2(\theta)$, i.e., with a privileged emission angle around 90° from the incident photon).

These fundamental characteristics enable to describe photoneutrons at the time of their creation and at the atomic nucleus scale. However, from a macroscopic point of view, several parameters of the electron accelerator influence the energy spectrum and angular distribution of the photoneutron fluxes generated, such as accelerator geometry, materials present and electron energy. This justifies the common practice of using Monte Carlo transport codes to characterize photoneutron fluxes emitted by electron accelerators.

On another note, photonuclear reactions are not the only reactions at the origin of neutron emission by electron accelerators in the 4–20 MeV range. According to the *equivalent photon approximation*, electrons can produce *virtual photons* leading to electronuclear reactions. Thus, electroneutrons can be produced simultaneously with photoneutrons. However, the literature is rather poor regarding the measurement of (e , n) or (e , $e'n$) reaction cross-sections. Nevertheless, many authors reported that the electroneutron yield cross-sections are about 137 times smaller than the photoneutron yield cross-sections. This ratio was originally stated in 1984 in the NCRP report No. 79 (NCRP, 1984) and unfortunately postulated without demonstration. In this work, we have shown that this value seems to be only appropriate in materials like tantalum or tungsten for an energy close to 18 MeV. Overall, the contamination of photoneutron fluxes by electroneutrons increases with the accelerator's operating electron energy. We evaluated that the photoneutron fluxes emitted by tungsten or tantalum targets are likely to be polluted by electroneutrons with contamination levels ranging from approximately 0.01% at 10 MeV to 2% at 20 MeV.

Furthermore, when coupling an electron accelerator equipped with a tungsten or tantalum target with a secondary target made of beryllium or heavy water, the production of electroneutrons is foreseen to be limited to the primary target. In other words, the production of electroneutrons in a beryllium or heavy water secondary target would be nil. Regarding the energy spectra and angular distributions of electroneutrons produced by electron accelerators in the energy range studied, and given the current lack of scientific knowledge on these parameters, we recommend in the meantime assuming that electroneutron fluxes are characterized by the same energy spectra and angular distributions as photoneutron fluxes. Even when electronuclear reactions are implemented in Monte Carlo transport codes like GEANT4 or FLUKA, we advise using the codes sparingly and with a high level of vigilance when simulating electroneutron fluxes. Fundamental knowledge on energy spectra and angular distributions, as well as validated electronuclear data and extensive Monte Carlo code benchmarking are all cruelly lacking to judge as reliable whatever they are simulation results concerning electroneutron production from electron accelerators.

Using the MCNP6.2 Monte Carlo code, we evaluated the characteristics of photoneutron fluxes emitted by electron accelerators operating between 4 and 20 MeV. First, we characterized photoneutron fluxes emitted by tungsten or tantalum targets irradiated by electrons in the

7–20 MeV range. Then, we characterized photoneutron fluxes emitted by heavy water or beryllium secondary targets coupled with electron accelerators operating in the 4–9 MeV range. The photoneutron fluxes were characterized according to three main parameters: photoneutron yield cross-sections, energy spectrum and angular distribution. With astonishment, we showed that (γ, xn) cross-section threshold errors present in MCNPX when using the ENDF library, discovered in 2012 for some tungsten isotopes, are unfortunately still relevant in 2022 for MCNP6.2. In particular, we demonstrate that erroneous parameters in the nuclear data of MCNP6.2 lead to (γ, xn) cross-section threshold errors for two tungsten isotopes, i.e., ^{182}W and ^{186}W , which leads to a global underestimation of photoneutron production in tungsten. For calculations involving a tungsten target irradiated by electrons between 7 and 20 MeV, we determined new scaling factors valid with MCNP6.2 in order to circumvent these problems in the nuclear data of the code (see Table 2), and which can be reused by the reader.

Furthermore, by taking an in-depth look at nuclear data libraries, we showed that photoneutron yield cross-sections are sometimes poorly evaluated below 20 MeV. Indeed, the case of ^{184}W is distinctive and would by the way provide additional evidence to support the undervaluation trend of MCNP simulation regarding photoneutron production by a tungsten target in the energy range of interest. We can also regret the absence of experimental data regarding (γ, xn) reaction cross-sections for ^{180}W and ^{183}W , which raises questions about the reliability of their evaluations. It should be recalled that tungsten is perhaps one of the most common photoneutron-producing material present in electron accelerators working in the 4–20 MeV range and thus concerns many Monte Carlo code users. On top of that, when (γ, xn) reaction cross-section threshold errors are absent in MCNP and experimental data are available, as it is the case for ^{181}Ta for instance, it is legitimate to ask whether more than 50 year-old – single – experimental evaluations should be reevaluated (and let's hope validated) or not.

In addition, this study also showed that the photoneutron energy spectrum simulated for a tantalum target has a non-physical shape under 0.5 MeV, which is perhaps due to erroneous algorithms in MCNP for photoneutron energy calculations. Moreover, this study enabled to infer that the MCNP code does not simulate the direct component of photoneutron fluxes for tungsten or tantalum targets with its specific angular distribution characterized by a predominant emission angle at 90° from the incident photon. Besides, while focusing on photoneutrons emitted by a heavy water secondary target and based on cross-sections measured by different authors, we brought to light that the maximum of the (γ, n) cross-section curve around 4.5 MeV for deuterium is undervalued in ENDF. In other words, the photoneutron flux emitted by a heavy water target coupled with an electron accelerator operating in the 4–9 MeV range is likely to be underestimated when simulated with the MCNP code. Regarding beryllium, in light of strong discrepancies between (γ, n) cross-sections for ^9Be taken from different references, we recommend conducting MCNP simulations involving a beryllium secondary target coupled with an electron accelerator operating in the 4–9 MeV range with the utmost attention given the dubious reliability of the (γ, xn) cross-sections for ^9Be implemented in MCNP6.2.

In view of assessing the performance of the MCNP code for the simulation of photoneutron fluxes delivered by electron accelerators in the energy range studied, we benchmarked MCNP against neutron activation measurements conducted with three different electron accelerators, and discussed the results and discrepancies obtained by referring to several articles published in the literature. Whatever the energy at which electron accelerators are operating in the 4–20 MeV range, photoneutron activation calculations based on Monte Carlo simulations conducted with the MCNP code systematically underestimate the experimental results. We showed that the underestimation climbs to a decade for photoneutrons produced in tungsten with a 9 MeV electron accelerator. When tungsten is the sole photoneutron-emitting material in the setup simulated and when a correction factor is used *a posteriori* to circumvent the (γ, xn) cross-section threshold errors in

MCNP, simulation results keep underestimating the experimental data by a few tens of percent in the 10–20 MeV range and the discrepancies are the same that the target is made of tungsten or tantalum. When it comes to photoneutrons emitted by a heavy water secondary target coupled with an electron accelerator operating between 4 and 9 MeV, simulation of neutron activation measurements carried with MCNP also underestimates experiment by a few tens of percent. For the case of medical accelerators, due to their complex geometries, the photonuclear reactions usually occur in different parts of the machine and on different materials (tungsten, lead, etc.), which renders non-trivial the application of a correction factor on simulation results during the post-processing step to compensate the (γ, xn) cross-section threshold errors for tungsten in MCNP. Operating a Saturne 43F medical accelerator at 12 MV, we showed that the underestimation of results when simulating neutron activation measurements are of the order of a factor of ten.

Regarding the MCNP code, whatever the version considered, among which: MCNP5, MCNPX, MCNP6.1, MCNP6.2, photoneutron kinematics modelling is not documented. We suspect that the same algorithms are used for photonuclear physics modelling in these different versions of the MCNP code. Another point is that the latter all provide photonuclear data at least from the ENDF library although slight differences can be found between certain versions. For instance, photonuclear cross-sections from CNDC are available in MCNPX but not in MCNP6. Anyway, the investigations carried out by Caro (2016) and Fynan (2020) enabled to identify with a high level of confidence the equations erroneously implemented in the MCNP code to model photoneutron physics. In fact, the MCNP code uses erroneous equations related to *inelastic neutron scattering* to describe photoneutron kinematics, which results in the overestimation of photoneutron energy and subsequently in significantly hardened photoneutron energy spectra. Consequently, because of the decrease of the (n, γ) cross-sections when the neutron energy increases, this spectrum-hardening effect would contribute to the underestimation of MCNP simulation results when simulating photoneutron activation measurements of vanadium or aluminium foils. However, inaccuracies of the photonuclear reaction cross-sections may also somehow exacerbate these discrepancies, in the context in which the reliability of photonuclear data has been openly discussed by Varlamov (2019).

6. Conclusions, recommendations and perspectives

In this paper, we critically reviewed the performance of Monte Carlo transport codes to simulate photoneutron fluxes emitted by electron accelerators operating in the 4–20 MeV range. In summary, from the three main sources of errors leading potentially to unreliable simulated photoneutron fluxes:

- incorrect implementation of certain nuclear data in Monte Carlo codes;
- invalid modeling of photonuclear physics in Monte Carlo codes;
- lack of fundamental knowledge of photoneutron yield cross-sections (both near the reaction energy thresholds and at the maximum of the cross-section curves);

several recommendations can be made. But first, let's underline that all improvements foreseen regarding photoneutron simulation accuracy should be done by keeping an eye on all of those three possible and cumulative sources of errors.

The continuous interest for photoneutrons from the accelerator community should convince Monte Carlo code development teams to seek to enhance simulation of photonuclear reactions. Both nuclear data files and photonuclear kinematics algorithms should be checked and improved. The feedbacks from code users could help to achieve these tasks in the most efficient way. We also encourage the developers to openly document and publish these upcoming enhancements. On

another note, the support of the experimental nuclear physics community could enable to re-evaluate photonuclear cross-sections and improve further nuclear data implemented in Monte Carlo codes, perhaps starting by working on the most common photoneutron-producing materials present in (or associated with) electron accelerators, i.e., tungsten, tantalum, lead, beryllium and deuterium.

Regarding the community of Monte Carlo code users, we suggest to avoid blind trust in simulation results involving the production of photoneutrons by electron accelerators in the 4–20 MeV range. Taking a step back and use of critical thinking is advisable when simulating photoneutrons, and even more in the case of electroneutrons. In practice, to estimate the level of reliability of simulation results involving photoneutrons, crosschecking with experimental results could represent a solution of interest, and if not possible, at least comparing results obtained with the ones provided by other codes. As further improvements in Monte Carlo simulation of photonuclear reactions may take some time to reach a high level of performance – and even more for the case of electroneutrons – it is wise for users to focus in the meantime on, first, understanding and then, mastering the current limitations of the codes.

Finally, considering the diversity of skills required, i.e., verification and validation or correction of photonuclear data in Monte Carlo transport codes, implementation of appropriate photonuclear physics algorithms as well as the establishment of new photonuclear cross-section evaluations, the key to such progress in photoneutron simulation in the coming years will certainly rely on international and collaborative work. Meanwhile, benchmarking several Monte Carlo transport codes developed by various teams could help to take another step forward with the objective of refining the validity domain of the different codes for the simulation of photoneutron fluxes emitted by electron accelerators operating in the 4–20 MeV range.

CRedit authorship contribution statement

Adrien Sari: Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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